COURSE SLIDES · WSQ

AI Vibe Coding for Augmented Reality (AR) Applications

WSQ Course Code: TGS-2022017519

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v9.1 · 12 September 2026

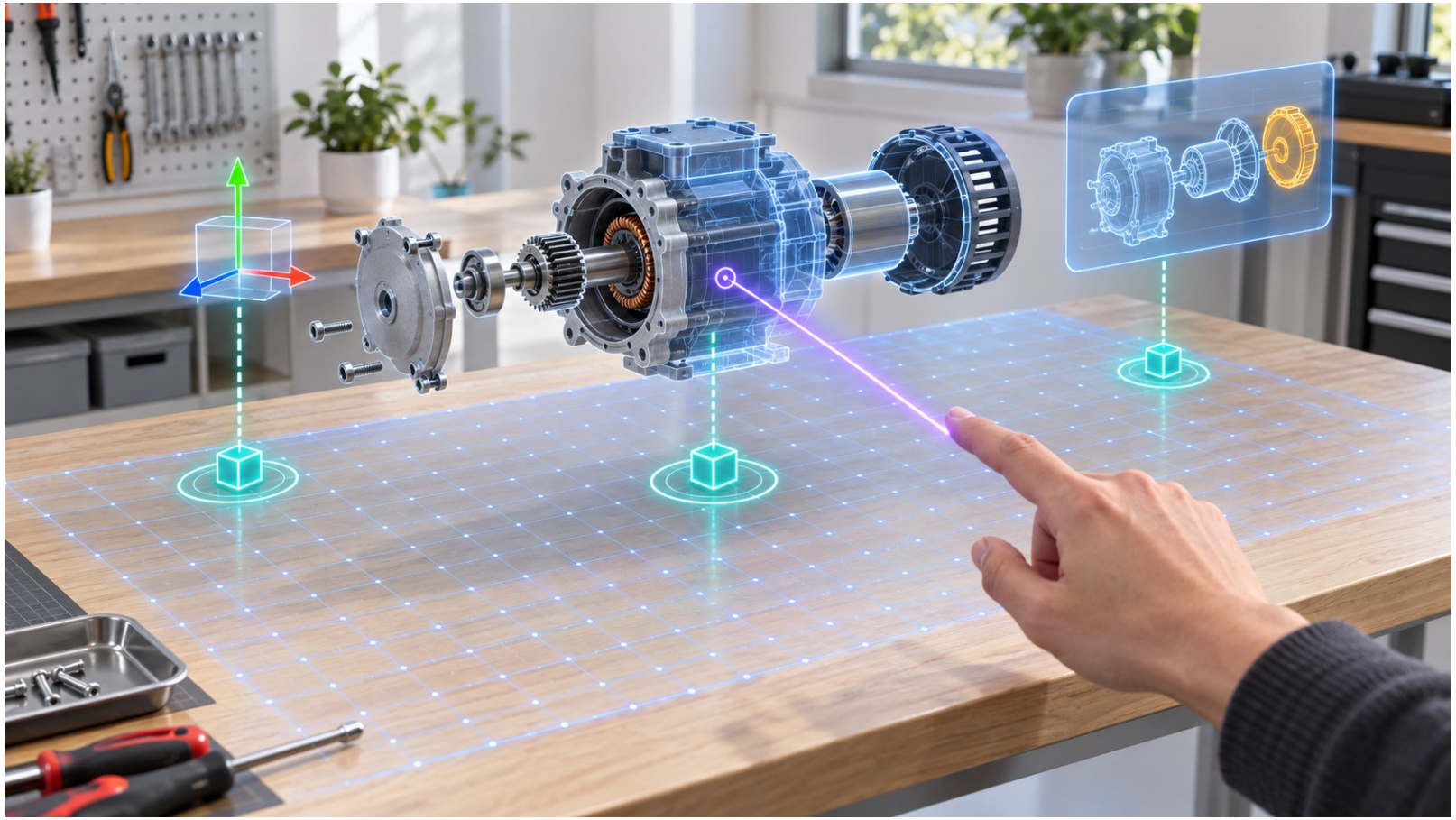
Build spatial experiences from natural language—then inspect the contract, runtime and evidence.



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MECHANISM · ANNOTATED VISUAL

Spatial computing, made inspectable



1 Detected plane

Real surface geometry becomes the placement constraint.

2 Session anchor

Virtual content follows a tracked pose for the active session.

3 Interaction ray

Input becomes a ray, nearest hit and visible feedback.

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Digital Attendance (Mandatory)

1 AM

Scan and submit the SSG QR code at the start of the morning block.

2 PM

Repeat attendance after lunch when the trainer displays the live code.

3 ASSESSMENT

Submit the assessment attendance event before the papers begin.

4 EVIDENCE

Minimum 75% attendance and competent assessment outcome apply.

COURSE RULE

Attendance is an official digital record; use the live QR displayed by the trainer.

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About the Trainer



GENERAL TRAINER PROFILE TEMPLATE

Complete before delivery

NAME

_____

QUALIFICATIONS

_____

EXPERTISE

_____

EXPERIENCE

_____

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About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE

Principal Trainer

QUALIFICATIONS

PhD · ACTA · DACE

TECHNICAL FOCUS

AI-assisted development and intelligent systems

FACILITATION

Mechanism-led demonstrations and supervised labs

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LEARNER INTRODUCTION

Let's know each other

1 **ROLE**

What work or learning context brings you here?

2 **BUILD**

What 3D, web or programming experience do you already have?

3 **DEVICE**

Which laptop, browser and optional XR device will you use?

4 **OUTCOME**

Which AR/VR workflow do you want to prototype?

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SAFE AND PRODUCTIVE LAB

Ground rules

1 PHYSICAL SAFETY

Clear the play area before immersive testing; remain aware of people and obstacles.

2 DATA SAFETY

Never paste credentials, learner data or private client material into a coding agent.

3 CODE REVIEW

Inspect every generated diff and preserve a rollback path.

4 EVIDENCE

A build is not proof; retain runtime state, checks and test output.

5 PARTICIPATION

Ask, test and explain—no question is too small.

6 DEVICES

Use silent mode; return from breaks on time.

7 PAIR REVIEW

Challenge assumptions respectfully and point to evidence.

8 ATTENDANCE

Minimum 75% attendance is required.

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Download course material

https://lms-tms.tertiaryinfotech.com/

Course materials are linked from the matching LMS course record

- Open the exact TGS-2022017519 course page.
- Confirm current version before downloading.
- Keep assessment answer keys trainer-only.

1 CHECK 1

Open the exact TGS-2022017519 course page.

2 CHECK 2

Confirm current version before downloading.

3 CHECK 3

Keep assessment answer keys trainer-only.

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Two-day lesson plan

1 **09:30**

Course/admin, learning outcomes and evidence boundaries

2 **10:15**

Topic 1 mechanisms + Labs 01–02

3 **13:30**

Lab 03 and Topic 2 scene/runtime mechanisms

4 **15:45**

Labs 04–05, runtime evidence and recap

COURSE RULE

Detailed procedures live in the Learner Guide and individual lab folders.

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DAY 2 · INTERACTION TO REVIEW

Two-day lesson plan

1 **09:30**

Lab 06 and interaction diagnostics

2 **11:30**

Topic 3 AR/VR boundaries + Lab 07

3 **13:30**

Performance, accessibility and Lab 08

4 **15:30**

Assessment briefing · WA 60 min · PP 90 min

COURSE RULE

The current Drive papers total 150 minutes; paper-level timings are retained.

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Skills Framework mapping

1 **K1–K3**

Storyboard · spatial UI · equipment specifications

2 **K4–K6**

Work procedure · AR runtime · real-world overlay

3 **K7**

Technical differences between VR and AR

4 **A1–A4**

Translate · storyboard · review · advise adjustments

COURSE RULE

Every code appears in the labs, guide and assessment mapping.

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COURSE OUTCOMES

Learning outcomes

1

LO1

Translate specifications and write storyboards for AR applications using spatial user-interface principles.

2

LO2

Apply AR working principles to create procedures and review AR runtimes for usability and clarity.

3

LO3

Overlay virtual objects on the real world, distinguish AR from VR, and advise developers on effective adjustments.

COURSE RULE

Success means producing inspectable AR/VR contracts, runtime evidence and developer advice—not prompt text alone.

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8 CONNECTED LABS · LIVE PLAYGROUND + LOCAL SOURCE

Lab roadmap

1 **LAB 01**

Scene Brief and Prompt Contract

2 **LAB 02**

Storyboard and Spatial UI

3 **LAB 03**

WebXR Capability and Equipment Gate

4 **LAB 04**

Scene Graph, Lighting and Animation

5 **LAB 05**

Plane Detection, Hit Test and Anchors

6 **LAB 06**

Raycast, Touch and Spatial Feedback

7 **LAB 07**

Immersive VR, Controllers and Comfort

8 **LAB 08**

Runtime Review, Optimisation and Handover

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ASSESSMENT PREPARATION

Briefing for Assessment

1 INDIVIDUAL

Complete both instruments independently.

2 OPEN BOOK

Use slides, Learner Guide and approved class materials.

3 SUBMIT

Complete the document provided and upload through the LMS.

4 EVIDENCE

PP answers cite lab artifacts and runtime review evidence.

COURSE RULE

Briefing occurs before the assessment begins.

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Final assessment

1 WA · 60 MIN

7 open-ended questions covering K1–K7.

2 PP · 90 MIN

3 connected tasks: A1+A2, A3, and A4.

3 COMPETENCY

Every required knowledge and ability criterion must be competent.

4 APPEAL

Discuss the decision, record intent and follow the documented appeal process.

COURSE RULE

The current papers total 150 minutes; this conflicts with the generic 2-hour course-page label and is retained to mirror the approved paper-level evidence.

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Practice and revision portal

https://exams.tertiaryinfotech.com/

Tertiary Exams

- Use the course self-checks and lab verifiers for this WSQ course.
- The public exam portal currently has no verified AR course-specific listing.
- Do not invent or substitute an unrelated certification exam.

1 **CHECK 1**

Use the course self-checks and lab verifiers for this WSQ course.

2 **CHECK 2**

The public exam portal currently has no verified AR course-specific listing.

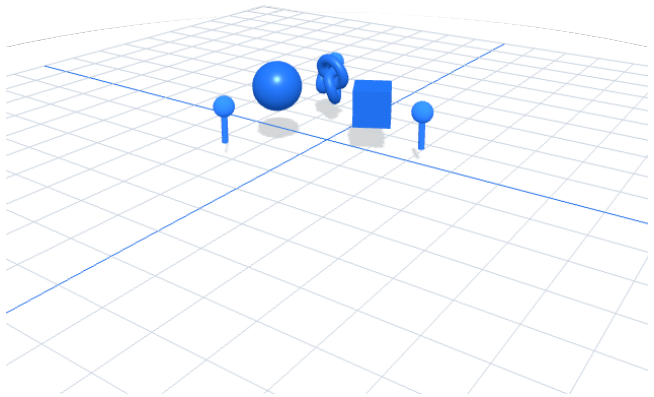
3 **CHECK 3**

Do not invent or substitute an unrelated certification exam.

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Try the live AR/VR Playground

https://alfredang.github.io/arvr-playground/



Prompt → contract → runtime → checks

- Open the live static HTML/CSS/JavaScript playground—no install required.
- Desktop preview works when immersive WebXR is unsupported.
- Use the localhost source when a lab asks you to edit code.

1 CHECK 1

Open the live static HTML/CSS/JavaScript playground—no install required.

2 CHECK 2

Desktop preview works when immersive WebXR is unsupported.

3 CHECK 3

Use the localhost source when a lab asks you to edit code.

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01

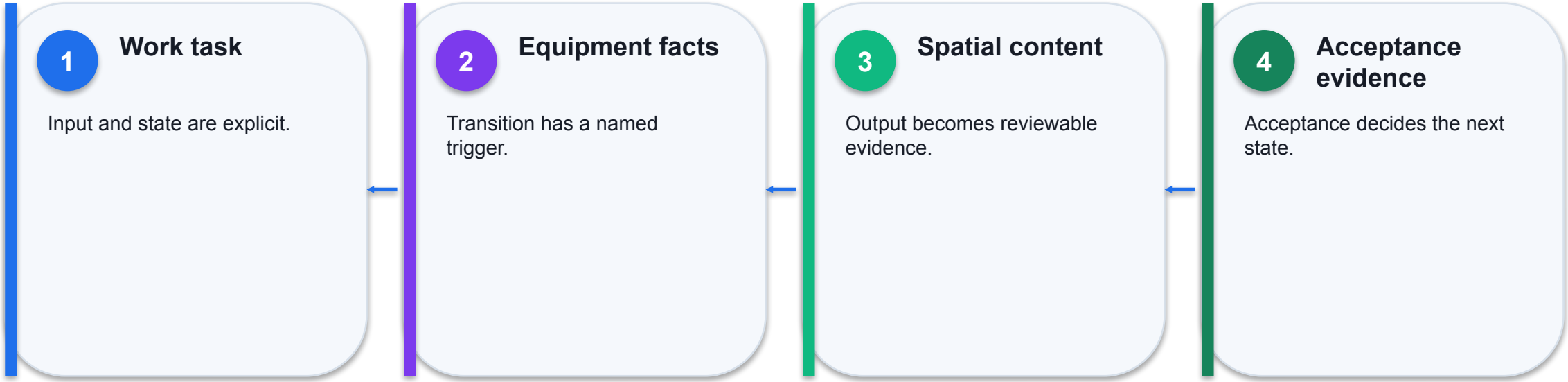
TOPIC 1 · K1 · K2 · K3 · A1 · A2

AR User Interface Design and Storyboarding with AI Vibe Coding

Requirements · prompt contracts · storyboard states · spatial UI · capability gates

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Requirement to scene contract



MECHANISM RULE

Every request becomes explicit targetMode, tracking, metric scale, content, interaction, performance and accessibility fields.

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Requirement to scene contract: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
const scene = parsePrompt(requirement, modeOverride);

// acceptance
A reviewer can point from each requirement to one JSON field and one visible runtime state.
```

INPUT	Work task
STATE	Equipment facts
CONTROL	Spatial content
EVIDENCE	Acceptance evidence

ACCEPTANCE — **A reviewer can point from each requirement to one JSON field and one visible runtime state.**

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Requirement to scene contract: failure and control

! FAILURE MODE

A vague prompt produces an attractive demo that cannot be assessed or safely handed over.

✓ CONTROL / RECOVERY

Require a versioned schema and reject any generated change that cannot be traced to one field.

VERIFICATION EVIDENCE

A reviewer can point from each requirement to one JSON field and one visible runtime state.

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Data boundary for coding agents



MECHANISM RULE

Prompts may include public specifications and local source excerpts; credentials, learner records and proprietary data remain outside the agent context.

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Data boundary for coding agents: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
git diff -- webapp/src && npm test

// acceptance
The prompt and diff contain no secret pattern, personal data or unrelated file change.
```

INPUT	**Classify input**
STATE	**Remove secrets**
CONTROL	**Minimise context**
EVIDENCE	**Inspect output**

ACCEPTANCE — **The prompt and diff contain no secret pattern, personal data or unrelated file change.**

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TOPIC 1 · DIAGNOSTIC

Data boundary for coding agents: failure and control

FAILURE MODE

A prompt includes an API key or personal record and the resulting transcript becomes a new disclosure surface.

CONTROL / RECOVERY

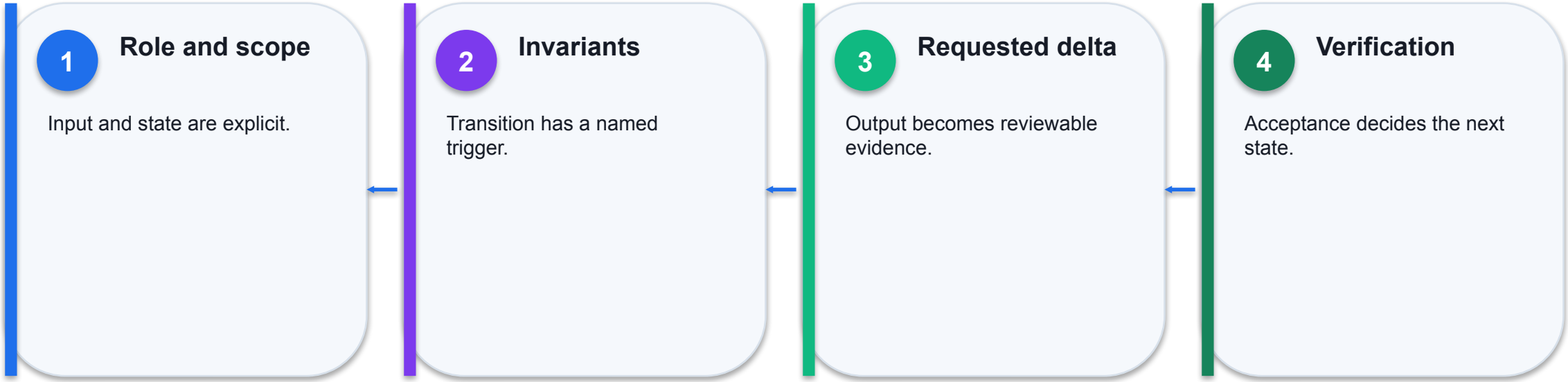
Use `.env.example`, synthetic data and named-file diffs; scan before commit.

VERIFICATION EVIDENCE

The prompt and diff contain no secret pattern, personal data or unrelated file change.

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Prompt contract anatomy



MECHANISM RULE

A bounded coding prompt states the repository, files to inspect, behaviour to preserve, smallest intended change and exact tests.

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Prompt contract anatomy: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
Read first. Do not edit yet. Return plan, risks, files and tests.

// acceptance
The response names unverified assumptions and the commands actually executed.
```

INPUT — Role and scope

STATE — Invariants

CONTROL — Requested delta

EVIDENCE — Verification

ACCEPTANCE — The response names unverified assumptions and the commands actually executed.

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Prompt contract anatomy: failure and control

!

FAILURE MODE

The agent rewrites working code, changes dependencies and claims success from syntax alone.

✓

CONTROL / RECOVERY

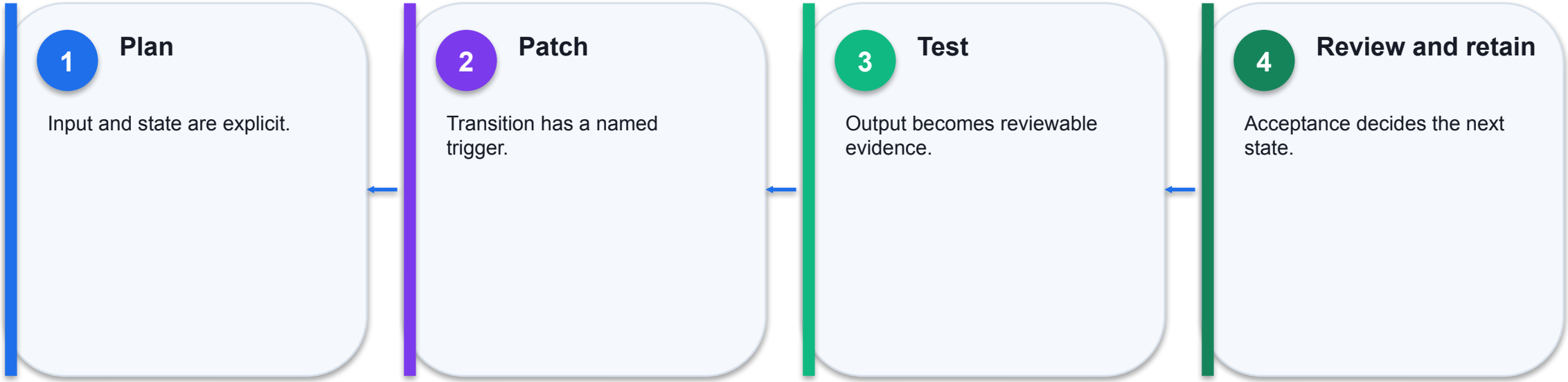
Separate plan, implementation, critique and verification prompts; inspect the diff after each.

VERIFICATION EVIDENCE

The response names unverified assumptions and the commands actually executed.

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Supervised vibe-coding loop



MECHANISM RULE

Human acceptance sits between every phase; generated code is a proposal until its behaviour and evidence pass.

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Supervised vibe-coding loop: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
npm audit --audit-level=moderate && npm test && npm run build

// acceptance
Plan matches diff; tests match acceptance criteria; runtime state matches the contract.
```

INPUT	Plan
STATE	Patch
CONTROL	Test
EVIDENCE	Review and retain

ACCEPTANCE — Plan matches diff; tests match acceptance criteria; runtime state matches the contract.

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TOPIC 1 · DIAGNOSTIC

Supervised vibe-coding loop: failure and control

!

FAILURE MODE

A green build hides a wrong spatial contract, inaccessible control or unsupported-device dead end.

✓

CONTROL / RECOVERY

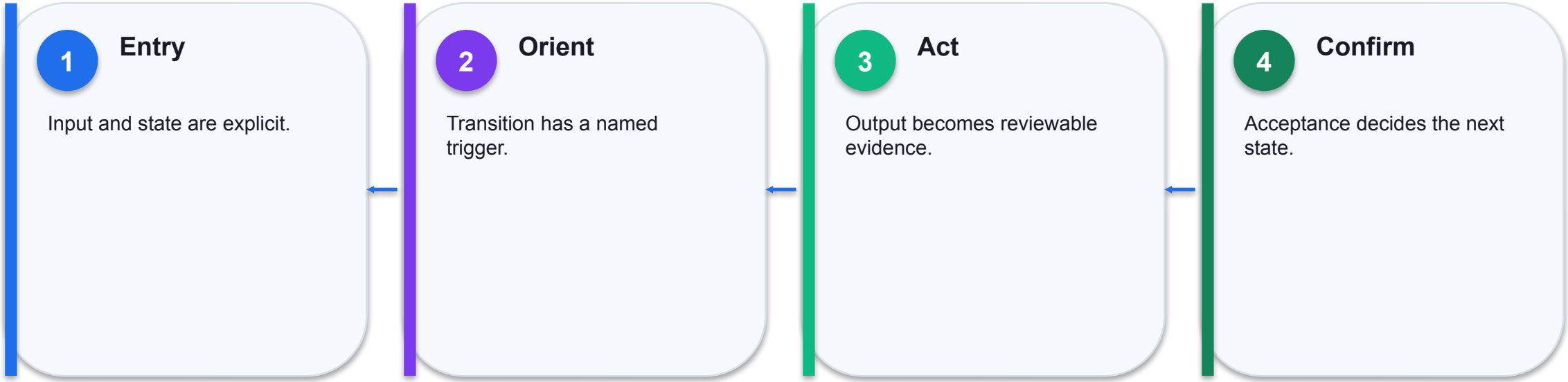
Combine automated tests with runtime checks, console/network evidence and device capability truth.

VERIFICATION EVIDENCE

Plan matches diff; tests match acceptance criteria; runtime state matches the contract.

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Storyboard as a state machine



MECHANISM RULE

Each frame names system state, user action and observable evidence—not merely a picture.

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Storyboard as a state machine: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
[{state:'Entry'}, {state:'Orient'}, {state:'Act'}, {state:'Confirm'}]

// acceptance
Every transition has a trigger and visible result that can be tested.
```

INPUT	Entry
STATE	Orient
CONTROL	Act
EVIDENCE	Confirm

ACCEPTANCE — **Every transition has a trigger and visible result that can be tested.**

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Storyboard as a state machine: failure and control

FAILURE MODE

Four attractive screens omit plane detection, selection feedback or completion state.

CONTROL / RECOVERY

Make transitions conditional: unsupported → fallback; no plane → scan; hit → select; complete → confirm.

VERIFICATION EVIDENCE

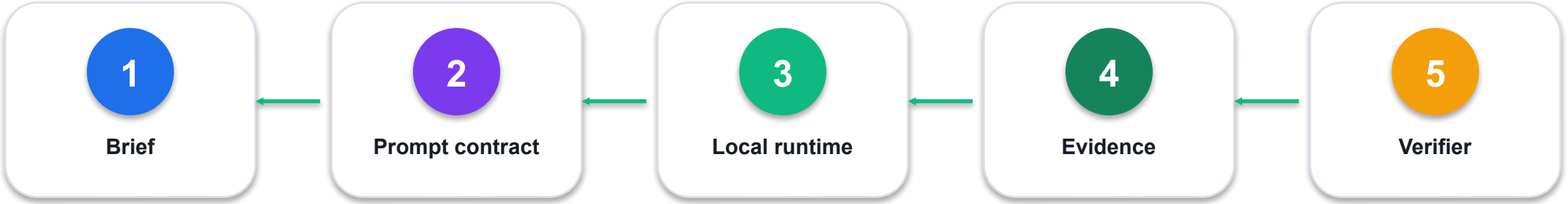
Every transition has a trigger and visible result that can be tested.

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TOPIC 1 · HANDS-ON LAB · 60 MIN

Lab 01 · Scene Brief and Prompt Contract

Translate a workplace requirement and equipment specification into a constrained, testable WebXR scene contract.

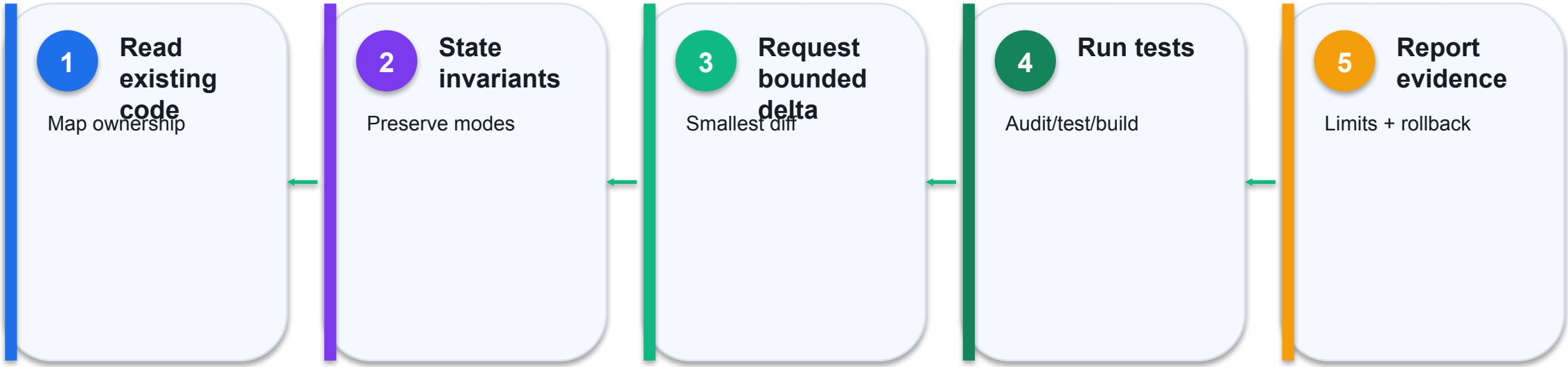


YOU'LL PRODUCE

Scene JSON · runtime evidence · review artifact · PASS Lab 01. Full detailed procedure: labs/lab-01-scene-brief-and-prompt-contract/README.md

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Lab 01: prompt-to-runtime contract



GUARDRAIL

Generated code remains a proposal until the JSON contract, runtime state and verifier agree.

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Lab 01: technical excerpt

```
{
  "targetMode": "ar",
  "tracking": {
    "plane": "horizontal",
    "placement": "hit-test",
    "anchorPersistence": "session"
  },
  "content": {
    "subject": "turbine",
    "instances": 3,
    "callouts": 3,
    "scaleMetres": 0.65
  },
  "performance": {
    "targetFps": 60,
    "drawCallBudget": 40,
    "triangleBudget": 80000
  }
}
```

1 **ASSERT 1** schema is ar-vr-vibe-scene/v1

2 **ASSERT 2** target mode is AR

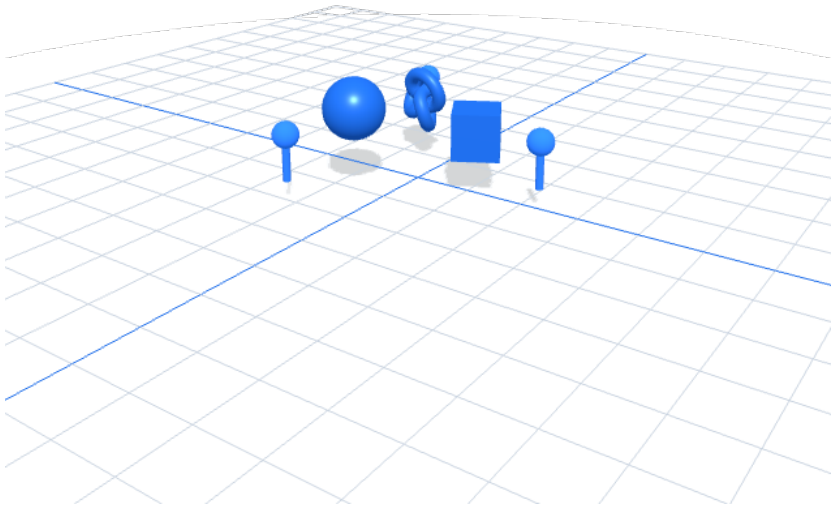
3 **ASSERT 3** subject is turbine

4 **ASSERT 4** three callouts are declared

RUN **node labs/lab-01-scene-brief-and-prompt-contract/verify.mjs → PASS Lab 01**

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Lab 01: acceptance evidence



PASS EVIDENCE

✓ schema is ar-vr-vibe-scene/v1

✓ target mode is AR

✓ subject is turbine

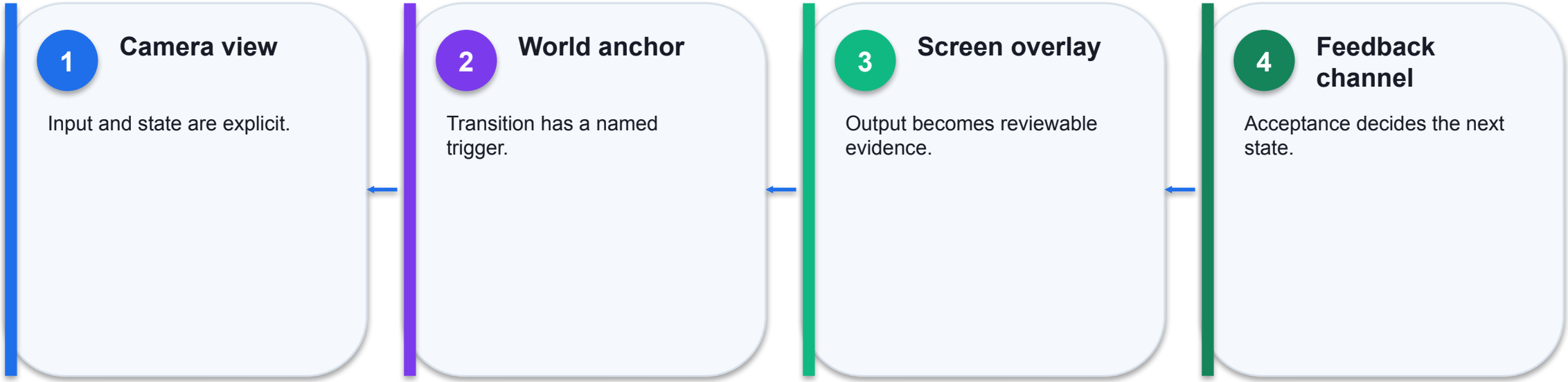
✓ three callouts are declared

DIAGNOSE

- If npm is missing, install the current Node.js LTS release and reopen Terminal.
- If the viewport is blank, reload once and check the browser console for WebGL errors.
- If the inferred mode is wrong, make `AR` explicit or choose the AR override.

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World-space versus screen-space UI



MECHANISM RULE

World-space labels move with an object; screen-space controls remain stable relative to the viewport.

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World-space versus screen-space UI: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
label.lookAt(camera.position);

// acceptance
Rotate the camera and verify labels retain context while controls remain reachable.
```

INPUT	Camera view
STATE	World anchor
CONTROL	Screen overlay
EVIDENCE	Feedback channel

ACCEPTANCE Rotate the camera and verify labels retain context while controls remain reachable.

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TOPIC 1 · DIAGNOSTIC

World-space versus screen-space UI: failure and control

FAILURE MODE

A critical control drifts out of reach or a label occludes the real hazard it is meant to explain.

CONTROL / RECOVERY

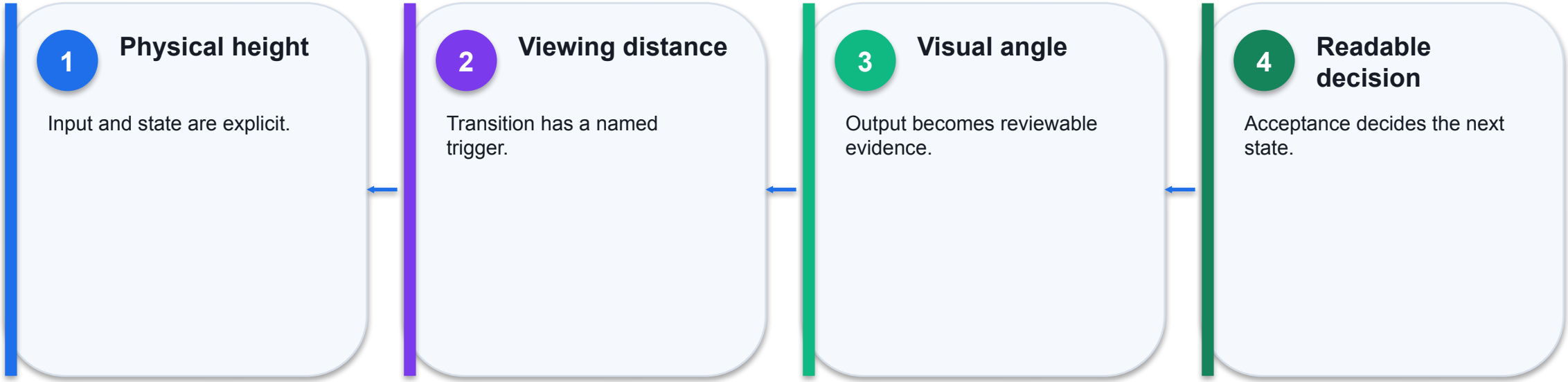
Choose the coordinate space from the information lifecycle; keep safety actions screen-stable.

VERIFICATION EVIDENCE

Rotate the camera and verify labels retain context while controls remain reachable.

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Angular size and viewing distance



MECHANISM RULE

Approximate angular size in radians as height ÷ distance for small angles; increasing distance halves apparent size.

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Angular size and viewing distance: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
angleRad = labelHeightMetres / distanceMetres

// acceptance
A 4 cm label at 1 m produces about 0.04 rad (2.3°) and remains within the tested range.
```

INPUT	Physical height
STATE	Viewing distance
CONTROL	Visual angle
EVIDENCE	Readable decision

ACCEPTANCE

A 4 cm label at 1 m produces about 0.04 rad (2.3°) and remains within the tested range.

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TOPIC 1 · DIAGNOSTIC

Angular size and viewing distance: failure and control

FAILURE MODE

A label that looks readable on a laptop becomes illegible at arm's length in AR.

CONTROL / RECOVERY

Test at minimum/typical/maximum distance and set scale or distance constraints in the contract.

VERIFICATION EVIDENCE

A 4 cm label at 1 m produces about 0.04 rad (2.3°) and remains within the tested range.

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Contrast over a live camera feed

1 **Sample background**

Input and state are explicit.

2 **Measure luminance**

Transition has a named trigger.

3 **Select surface**

Output becomes reviewable evidence.

4 **Retest in motion**

Acceptance decides the next state.

MECHANISM RULE

Camera luminance changes continuously; a label needs an opaque/semi-opaque plate or outline, not a fixed text colour alone.

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Contrast over a live camera feed: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
material.depthTest = false; labelBackdrop.opacity = 0.86;

// acceptance
Labels remain readable over light and dark backgrounds without colour alone.
```

INPUT	Sample background
STATE	Measure luminance
CONTROL	Select surface
EVIDENCE	Retest in motion

ACCEPTANCE Labels remain readable over light and dark backgrounds without colour alone.

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TOPIC 1 · DIAGNOSTIC

Contrast over a live camera feed: failure and control

FAILURE MODE

White text disappears over a bright wall; dark text disappears over machinery.

CONTROL / RECOVERY

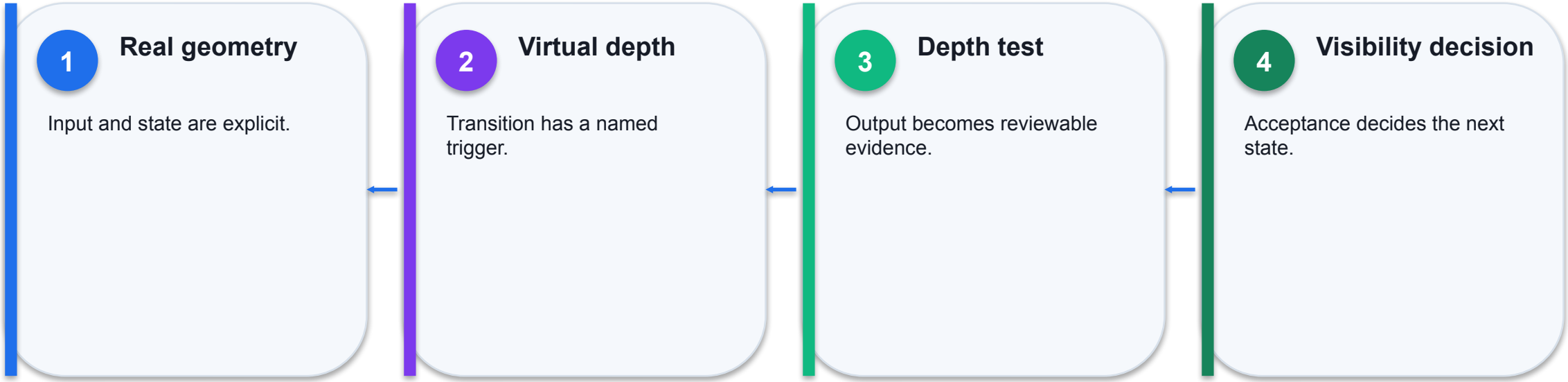
Use a high-contrast plate, redundant icon/shape cues and real-environment testing.

VERIFICATION EVIDENCE

Labels remain readable over light and dark backgrounds without colour alone.

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Depth ordering and occlusion



MECHANISM RULE

The depth buffer orders virtual fragments; real-world occlusion additionally requires depth/mesh data when supported.

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Depth ordering and occlusion: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
material.depthTest = true; material.depthWrite = true;

// acceptance
Near virtual surfaces occlude far ones; unsupported real-depth does not produce false claims.
```

INPUT	Real geometry
STATE	Virtual depth
CONTROL	Depth test
EVIDENCE	Visibility decision

ACCEPTANCE — **Near virtual surfaces occlude far ones; unsupported real-depth does not produce false claims.**

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TOPIC 1 · DIAGNOSTIC

Depth ordering and occlusion: failure and control

! FAILURE MODE

A virtual part draws through another object or hides the physical control surface.

✓ CONTROL / RECOVERY

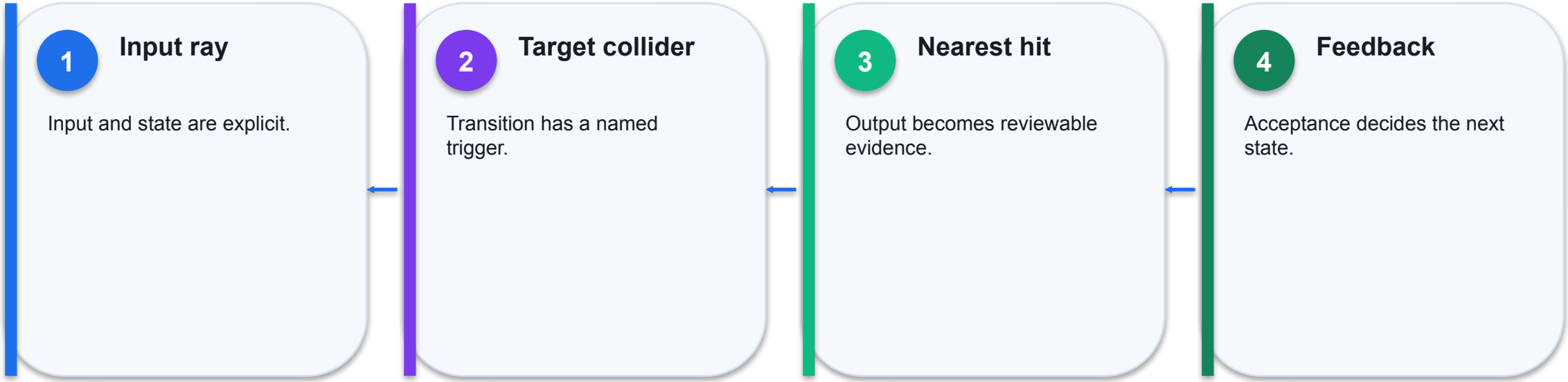
Declare the occlusion assumption and provide a non-occluding fallback when environment depth is unavailable.

VERIFICATION EVIDENCE

Near virtual surfaces occlude far ones; unsupported real-depth does not produce false claims.

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Reachability and target bounds



MECHANISM RULE

Interaction uses a ray origin/direction and ordered intersections; visible size and collider size may differ.

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Reachability and target bounds: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
const hit = raycaster.intersectObjects(selectables, true)[0];

// acceptance
Three target distances return one nearest valid hit; empty space returns a visible no-hit state.
```

INPUT	Input ray
STATE	Target collider
CONTROL	Nearest hit
EVIDENCE	Feedback

ACCEPTANCE Three target distances return one nearest valid hit; empty space returns a visible no-hit state.

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TOPIC 1 · DIAGNOSTIC

Reachability and target bounds: failure and control

FAILURE MODE

The visible object is selectable only in a tiny area or the far object wins through an overlapping target.

CONTROL / RECOVERY

Use forgiving invisible hit geometry, nearest-valid-hit rules and immediate feedback.

VERIFICATION EVIDENCE

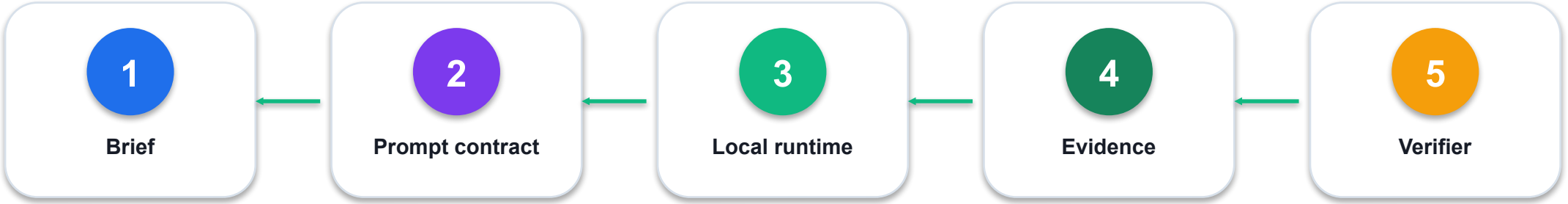
Three target distances return one nearest valid hit; empty space returns a visible no-hit state.

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TOPIC 1 · HANDS-ON LAB · 75 MIN

Lab 02 · Storyboard and Spatial UI

Convert an AR user journey into four observable states and apply readable, reachable spatial UI rules.

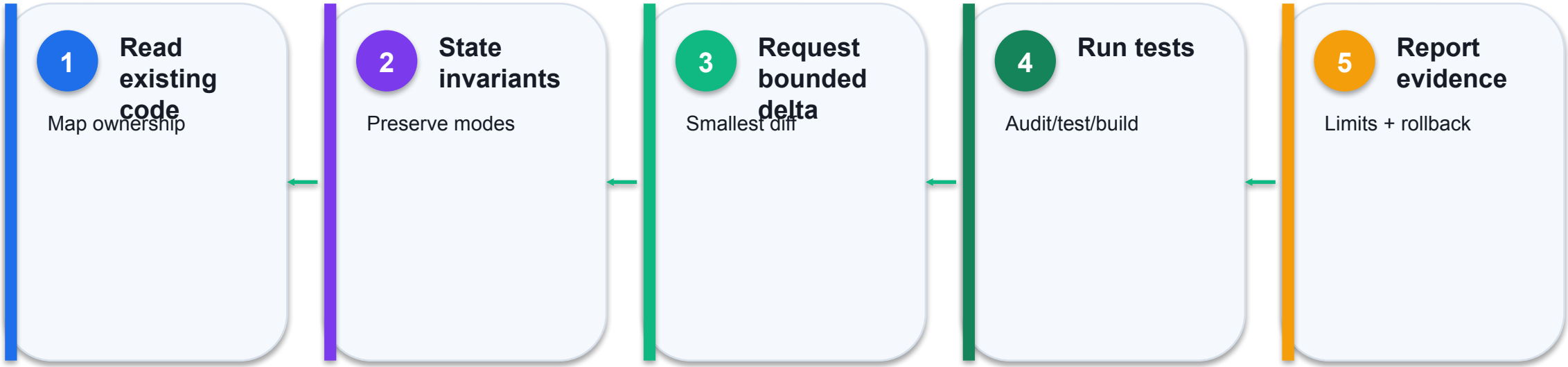


YOU'LL PRODUCE

Scene JSON · runtime evidence · review artifact · PASS Lab 02. Full detailed procedure: labs/lab-02-storyboard-and-spatial-ui/README.md

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Lab 02: prompt-to-runtime contract



GUARDRAIL

Generated code remains a proposal until the JSON contract, runtime state and verifier agree.

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Lab 02: technical excerpt

```
{
  "targetMode": "ar",
  "tracking": {
    "plane": "horizontal",
    "placement": "hit-test",
    "anchorPersistence": "session"
  },
  "content": {
    "subject": "machine",
    "instances": 4,
    "callouts": 4,
    "scaleMetres": 0.65
  },
  "performance": {
    "targetFps": 60,
    "drawCallBudget": 40,
    "triangleBudget": 80000
  }
}
```

1 **ASSERT 1** storyboard has four states

2 **ASSERT 2** high-contrast labels are enabled

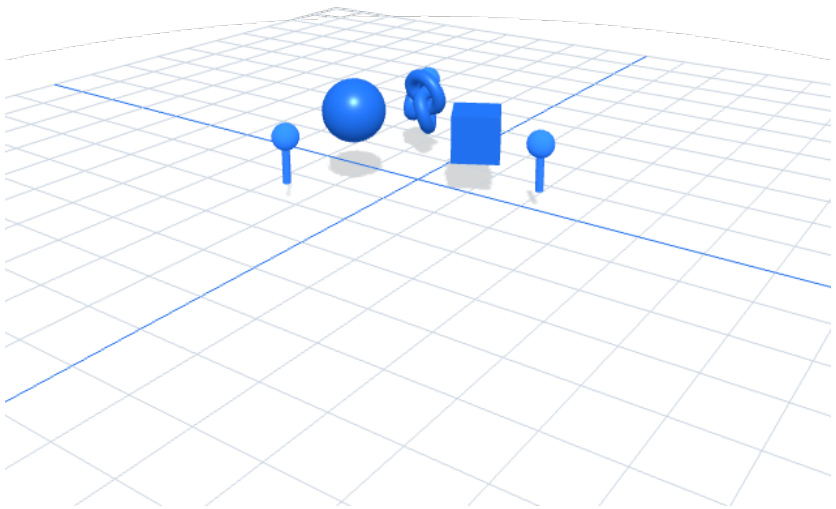
3 **ASSERT 3** reduced-motion control is present

4 **ASSERT 4** tap interaction is declared

RUN **node labs/lab-02-storyboard-and-spatial-ui/verify.mjs → PASS Lab 02**

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Lab 02: acceptance evidence



PASS EVIDENCE

✓ storyboard has four states

✓ high-contrast labels are enabled

✓ reduced-motion control is present

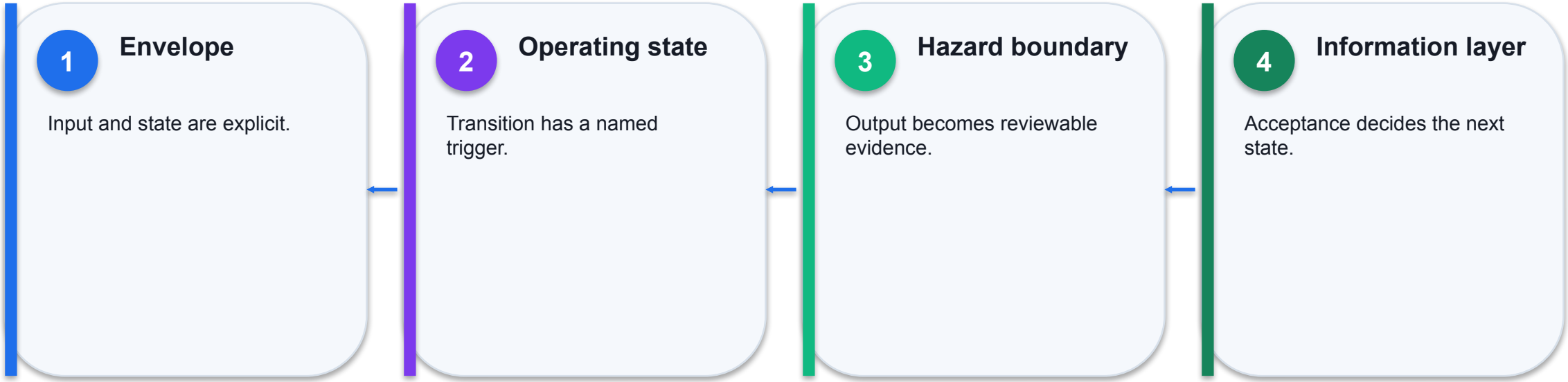
✓ tap interaction is declared

DIAGNOSE

- If a storyboard frame is vague, rewrite it as a visible state plus user action plus evidence.
- If labels disappear, enable Spatial callouts in Runtime controls.
- If mobile text is cramped, use the 390 px responsive layout and verify there is no horizontal overflow.

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Equipment specifications into AR content



MECHANISM RULE

Extract physical dimensions, safe operating state, controls that must remain visible and the exact instruction content.

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Equipment specifications into AR content: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
scaleMetres: 0.65, callouts: 3, anchorPersistence: 'session'

// acceptance
Measured scale and unobstructed safety controls are captured in the review evidence.
```

INPUT	Envelope
STATE	Operating state
CONTROL	Hazard boundary
EVIDENCE	Information layer

ACCEPTANCE — Measured scale and unobstructed safety controls are captured in the review evidence.

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Equipment specifications into AR content: failure and control

FAILURE MODE

A correctly modelled turbine is scaled wrongly or covers the emergency-stop control.

CONTROL / RECOVERY

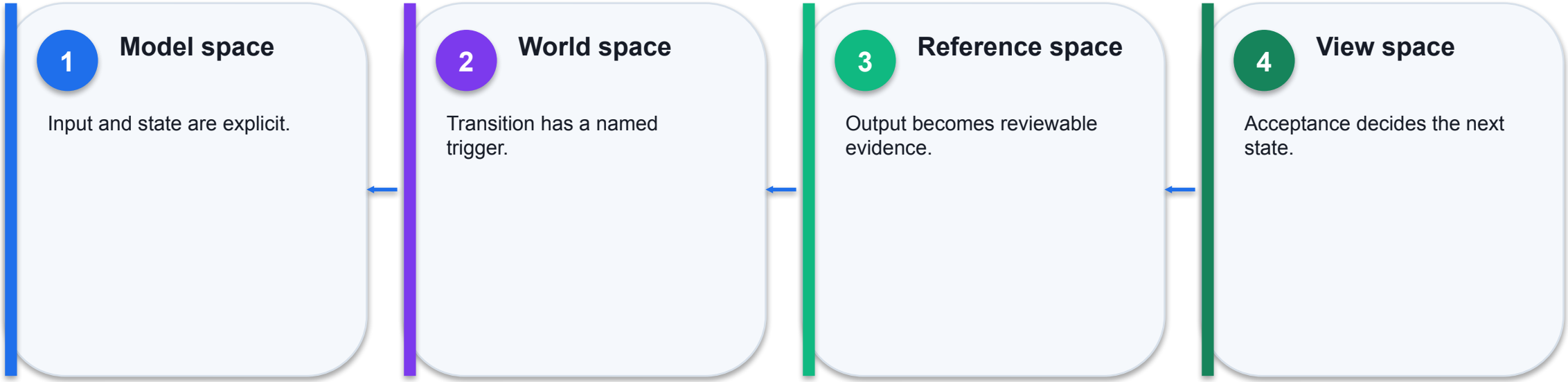
Treat physical envelope and no-occlusion zones as acceptance constraints.

VERIFICATION EVIDENCE

Measured scale and unobstructed safety controls are captured in the review evidence.

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Coordinate systems and metric units



MECHANISM RULE

Transforms compose from child to parent; WebXR reference spaces relate tracked poses to application world coordinates.

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Coordinate systems and metric units: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
worldMatrix = parent.matrixWorld * child.matrix

// acceptance
A 0.65 m source object measures 0.65 world units after import.
```

INPUT	Model space
STATE	World space
CONTROL	Reference space
EVIDENCE	View space

ACCEPTANCE — **A 0.65 m source object measures 0.65 world units after import.**

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TOPIC 1 · DIAGNOSTIC

Coordinate systems and metric units: failure and control

! FAILURE MODE

Centimetre-authored content interpreted as metres appears 100× too large.

✓ CONTROL / RECOVERY

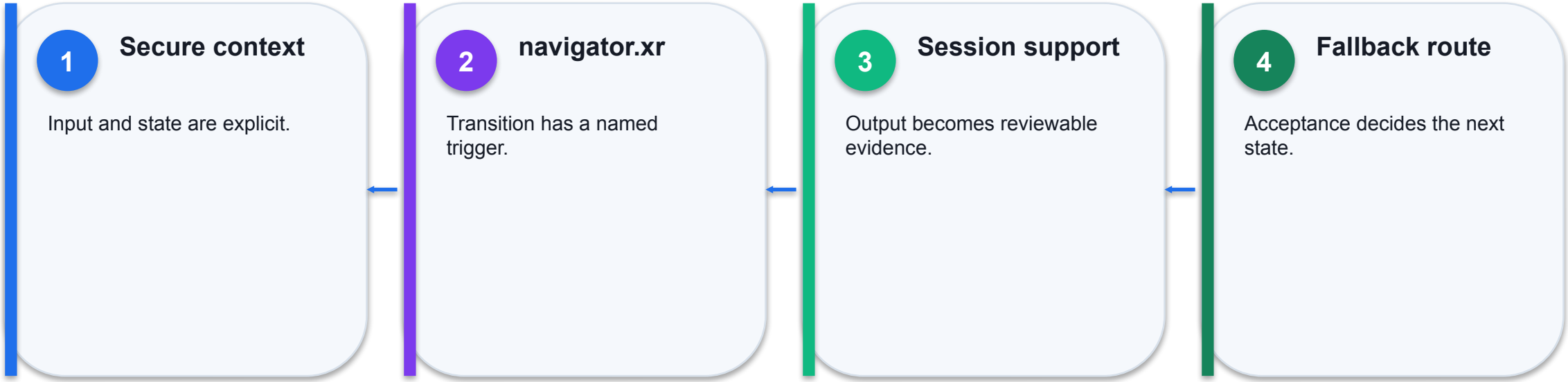
Normalise assets to metres at import and document the root scale.

VERIFICATION EVIDENCE

A 0.65 m source object measures 0.65 world units after import.

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Device capability decision matrix



MECHANISM RULE

Capability is measured, not inferred from device brand; immersive modes may be independently supported.

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Device capability decision matrix: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
await navigator.xr?.isSessionSupported('immersive-ar')

// acceptance
Unsupported AR/VR presents the exact limitation and a working preview route.
```

INPUT	Secure context
STATE	navigator.xr
CONTROL	Session support
EVIDENCE	Fallback route

ACCEPTANCE — **Unsupported AR/VR presents the exact limitation and a working preview route.**

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Device capability decision matrix: failure and control

FAILURE MODE

The AR button is shown on an unsupported desktop and ends in a dead click.

CONTROL / RECOVERY

Gate each mode separately and keep desktop preview available.

VERIFICATION EVIDENCE

Unsupported AR/VR presents the exact limitation and a working preview route.

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Secure context and permission intent



MECHANISM RULE

Immersive sessions require secure-context APIs and a user gesture; tracking should start only after explicit intent.

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Secure context and permission intent: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
button.addEventListener('click', () =>
navigator.xr.requestSession('immersive-ar'))

// acceptance
No tracking/session request occurs until the learner activates the
immersive button.
```

INPUT	Localhost or HTTPS
STATE	User activation
CONTROL	Session request
EVIDENCE	Consent duration

ACCEPTANCE — No tracking/session request occurs until the learner activates the immersive button.

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Secure context and permission intent: failure and control

FAILURE MODE

The app requests spatial tracking on page load or suggests bypassing a browser security warning.

CONTROL / RECOVERY

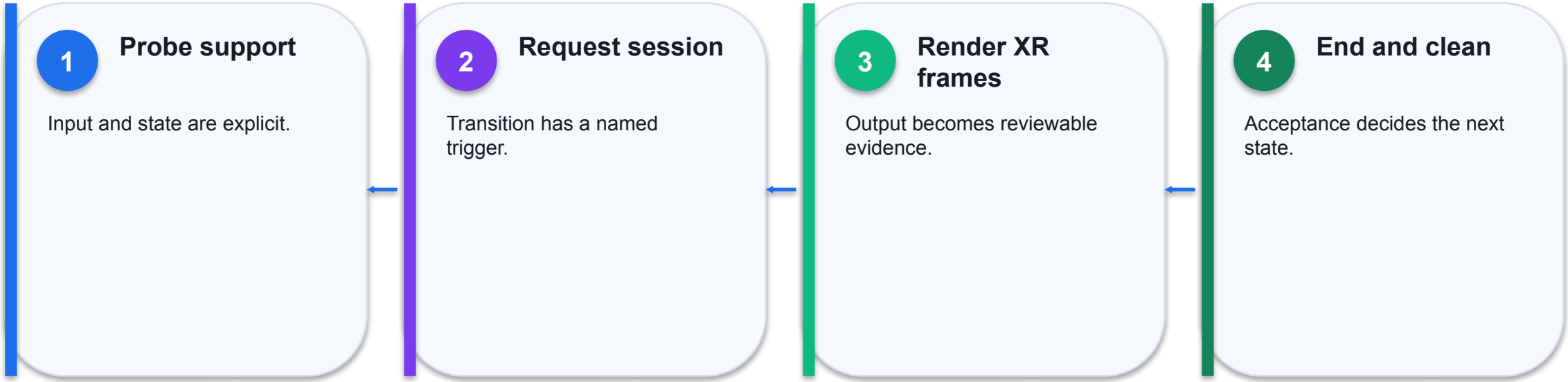
Use localhost/HTTPS and a clear Enter AR/VR action; never bypass safety interstitials.

VERIFICATION EVIDENCE

No tracking/session request occurs until the learner activates the immersive button.

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XR session lifecycle



MECHANISM RULE

The session owns frame timing, input sources and reference spaces; session-end cleanup returns the app to desktop state.

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XR session lifecycle: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
session.addEventListener('end', () => { hitTestSource = null; });

// acceptance
Start–end–restart produces one fresh source and no duplicated event handlers.
```

INPUT	Probe support
STATE	Request session
CONTROL	Render XR frames
EVIDENCE	End and clean

ACCEPTANCE Start–end–restart produces one fresh source and no duplicated event handlers.

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TOPIC 1 · DIAGNOSTIC

XR session lifecycle: failure and control

FAILURE MODE

A stale hit-test source or controller state leaks into the next session.

CONTROL / RECOVERY

Register one lifecycle owner and reset session-scoped resources on `end`.

VERIFICATION EVIDENCE

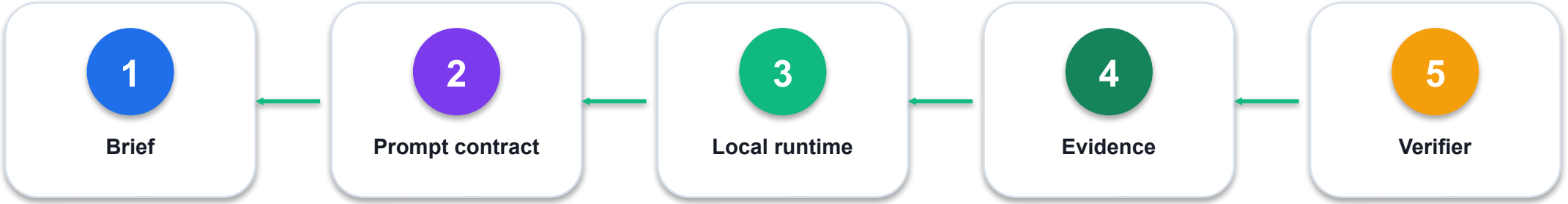
Start–end–restart produces one fresh source and no duplicated event handlers.

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TOPIC 1 · HANDS-ON LAB · 60 MIN

Lab 03 · WebXR Capability and Equipment Gate

Detect browser/device capability and route learners to AR, VR, or desktop fallback without a dead-end experience.

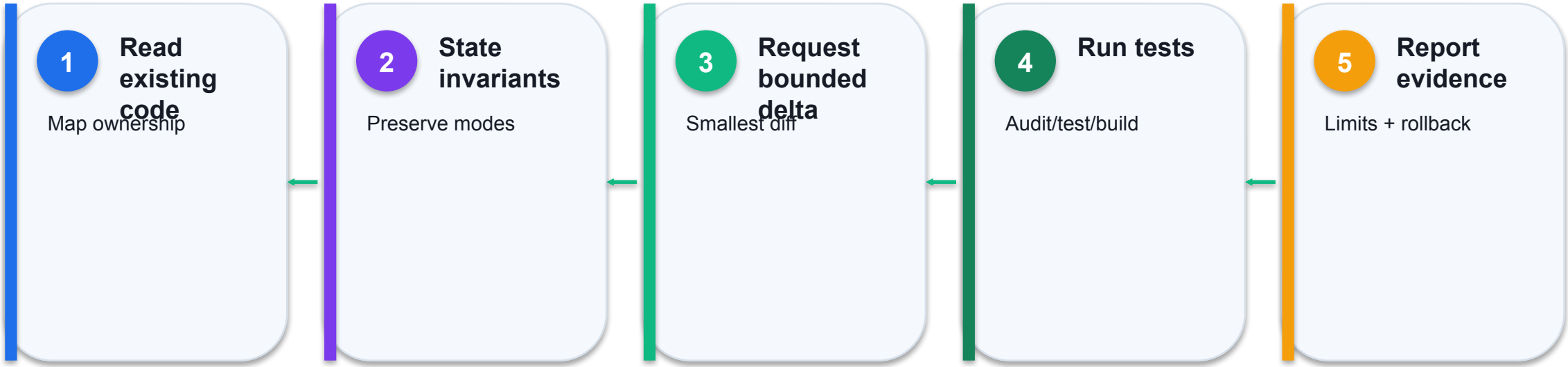


YOU'LL PRODUCE

Scene JSON · runtime evidence · review artifact · PASS Lab 03. Full detailed procedure: labs/lab-03-webxr-capability-and-equipment-gate/README.md

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Lab 03: prompt-to-runtime contract



GUARDRAIL

Generated code remains a proposal until the JSON contract, runtime state and verifier agree.

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Lab 03: technical excerpt

```
{
  "targetMode": "desktop",
  "tracking": {
    "orbitCamera": true
  },
  "content": {
    "subject": "product",
    "instances": 2,
    "callouts": 2,
    "scaleMetres": 0.65
  },
  "performance": {
    "targetFps": 60,
    "drawCallBudget": 40,
    "triangleBudget": 80000
  }
}
```

1 **ASSERT 1** desktop fallback remains available

2 **ASSERT 2** mode is explicit

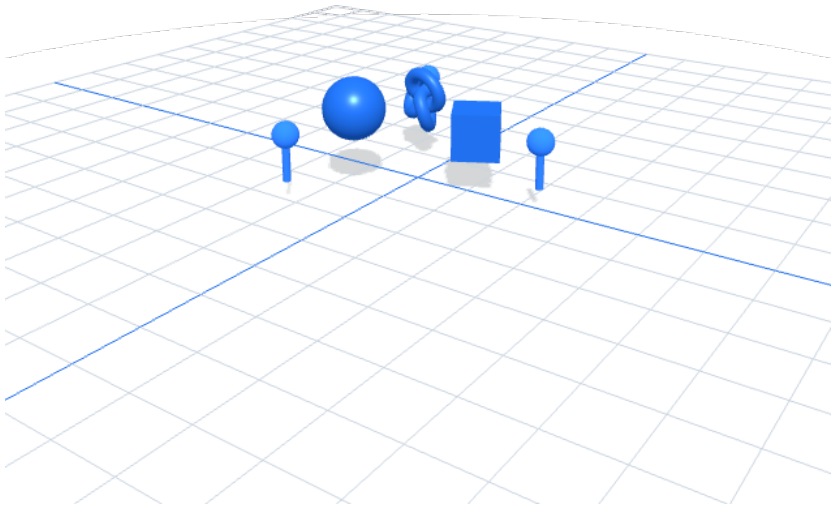
3 **ASSERT 3** runtime budget is declared

4 **ASSERT 4** unsupported WebXR is communicated

RUN **node labs/lab-03-webxr-capability-and-equipment-gate/verify.mjs → PASS Lab 03**

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Lab 03: acceptance evidence



PASS EVIDENCE

✓ desktop fallback remains available

✓ mode is explicit

✓ runtime budget is declared

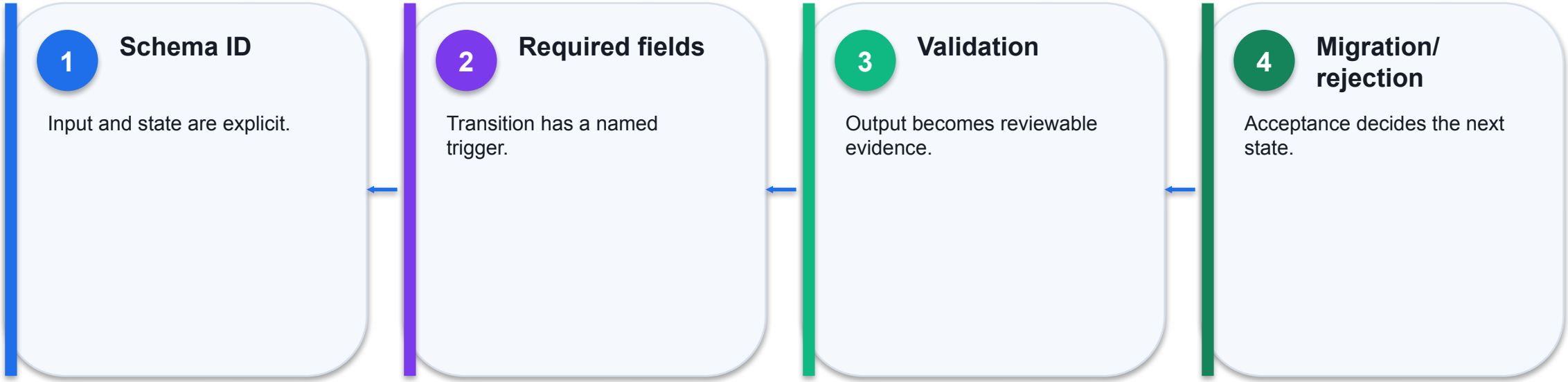
✓ unsupported WebXR is communicated

DIAGNOSE

- WebXR immersive sessions require HTTPS or localhost; use the exact local URL printed by Vite.
- Safari desktop may expose no immersive WebXR; use desktop preview for code review.
- If WebGL is disabled, confirm hardware acceleration before changing application code.

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Versioned scene schema



MECHANISM RULE

`ar-vr-vibe-scene/v1` is a contract; import rejects unknown schemas and invalid performance/accessibility fields.

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Versioned scene schema: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
if (scene.schema !== 'ar-vr-vibe-scene/v1') throw new Error('Unsupported scene schema');

// acceptance
Valid v1 imports round-trip; unknown schema fails before scene mutation.
```

INPUT	Schema ID
STATE	Required fields
CONTROL	Validation
EVIDENCE	Migration/rejection

ACCEPTANCE Valid v1 imports round-trip; unknown schema fails before scene mutation.

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Versioned scene schema: failure and control

FAILURE MODE

Silent import of an old or malformed scene fails later inside rendering code.

CONTROL / RECOVERY

Validate at the boundary and give a precise repair path.

VERIFICATION EVIDENCE

Valid v1 imports round-trip; unknown schema fails before scene mutation.

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Day 1 knowledge check

1 PROMPT

Which fields make a vibe prompt testable?

2 STORYBOARD

Which state proves the user can orient before acting?

3 CAPABILITY

Why are AR and VR probed independently?

4 EVIDENCE

What is the difference between a green build and runtime proof?

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02

TOPIC 2 · K4 · K5 · K6 · A3

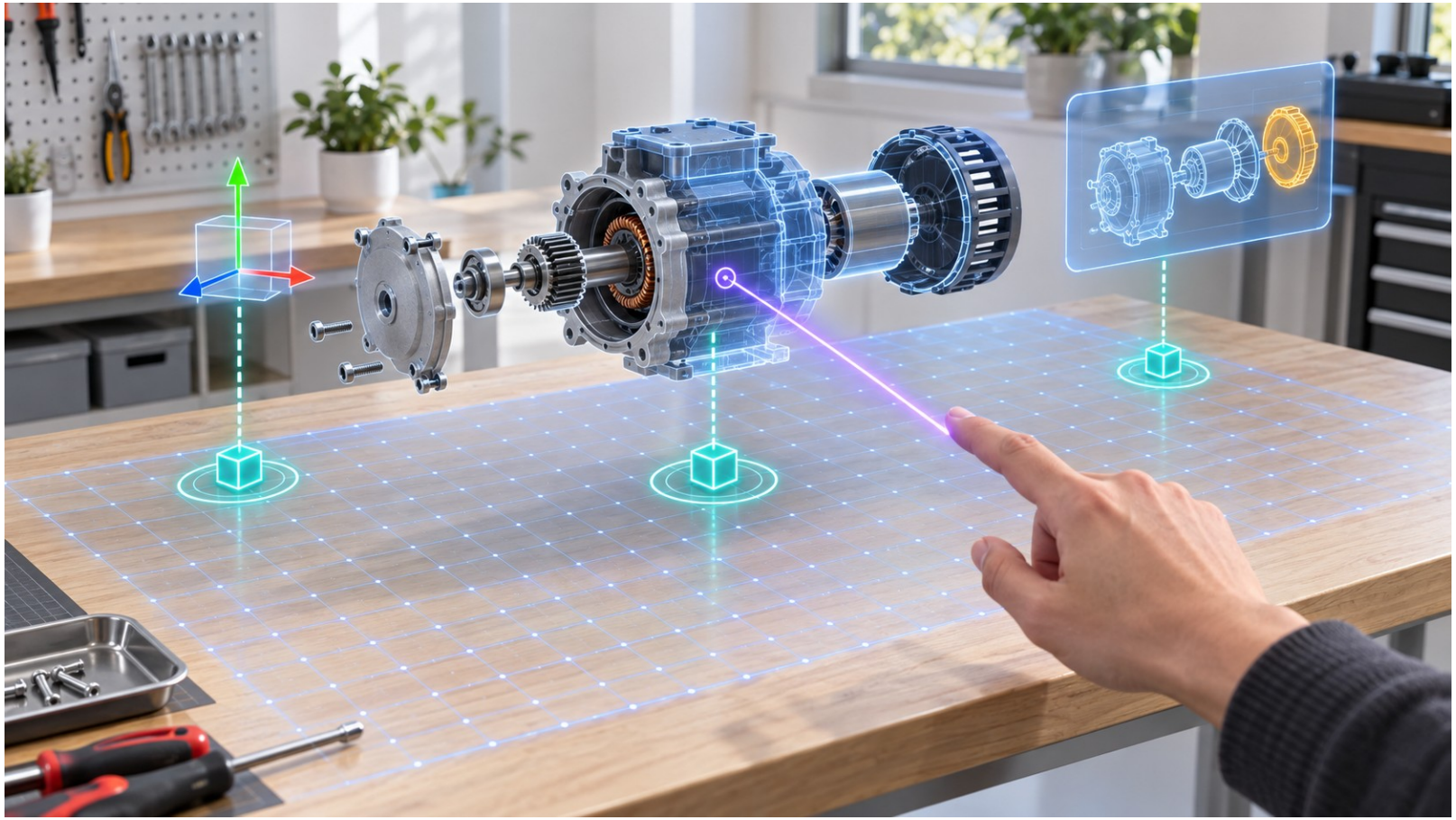
AR Application Development with Localhost WebXR

Three.js scene graph · WebXR sessions · hit test · anchors · raycast interaction

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MECHANISM · ANNOTATED VISUAL

Detected planes, hit tests and anchors



1 Plane

Hit testing queries real-world geometry exposed by the XR device.

2 Reticle

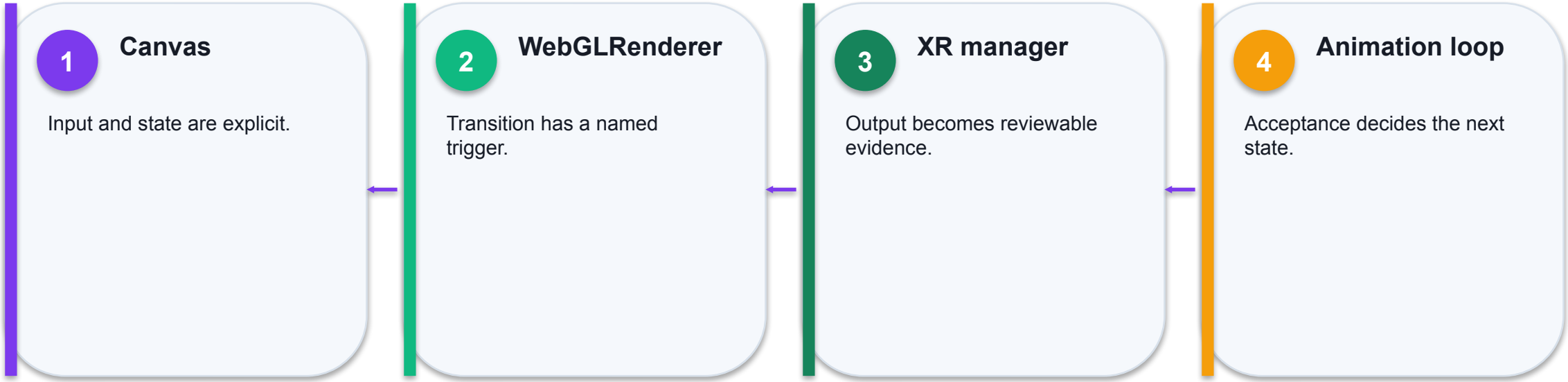
Current-frame pose is visible before the learner commits placement.

3 Anchor

Placed content inherits the session tracking boundary.

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Three.js renderer ownership



MECHANISM RULE

One renderer owns the drawing buffer, XR session binding, pixel ratio and render loop.

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Three.js renderer ownership: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
renderer.xr.enabled = true; renderer.setAnimationLoop(render);

// acceptance
Desktop and VR/AR modes render through one loop with no duplicate callback.
```

INPUT	Canvas
STATE	WebGLRenderer
CONTROL	XR manager
EVIDENCE	Animation loop

ACCEPTANCE Desktop and VR/AR modes render through one loop with no duplicate callback.

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TOPIC 2 · DIAGNOSTIC

Three.js renderer ownership: failure and control

! FAILURE MODE

Two animation loops compete or window.requestAnimationFrame stops during immersive presentation.

✓ CONTROL / RECOVERY

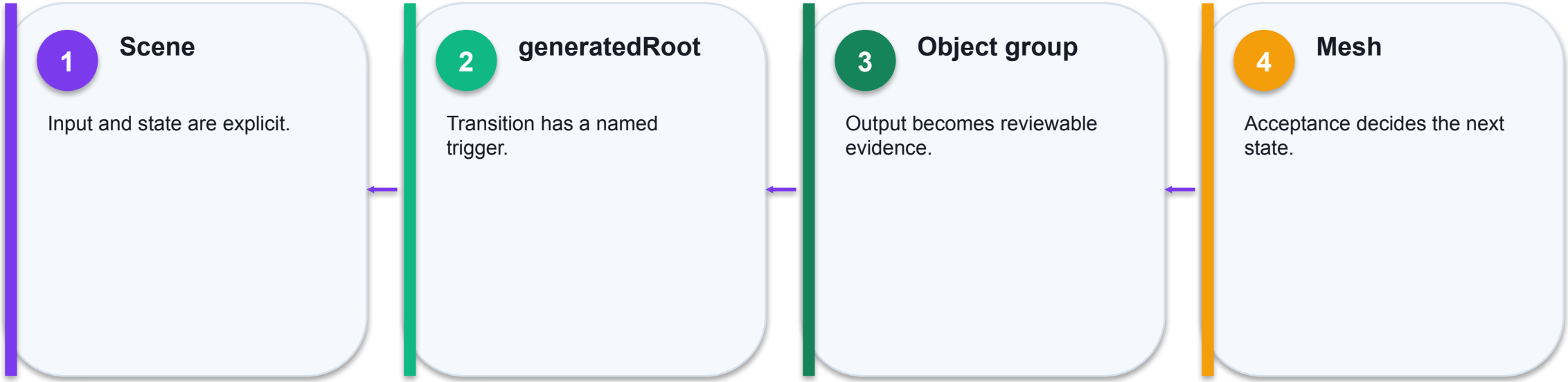
Use `setAnimationLoop` as the single frame scheduler for desktop and XR.

VERIFICATION EVIDENCE

Desktop and VR/AR modes render through one loop with no duplicate callback.

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Scene graph transform chain



MECHANISM RULE

A child transform is evaluated relative to its parent; root scale and placement move every generated object coherently.

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Scene graph transform chain: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
generatedRoot.position.setFromMatrixPosition(reticle.matrix);

// acceptance
The root follows placement while child motion preserves anchor locations.
```

INPUT	Scene
STATE	generatedRoot
CONTROL	Object group
EVIDENCE	Mesh

ACCEPTANCE — **The root follows placement while child motion preserves anchor locations.**

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TOPIC 2 · DIAGNOSTIC

Scene graph transform chain: failure and control

FAILURE MODE

Rotating the root also rotates callout anchors around the wrong pivot.

CONTROL / RECOVERY

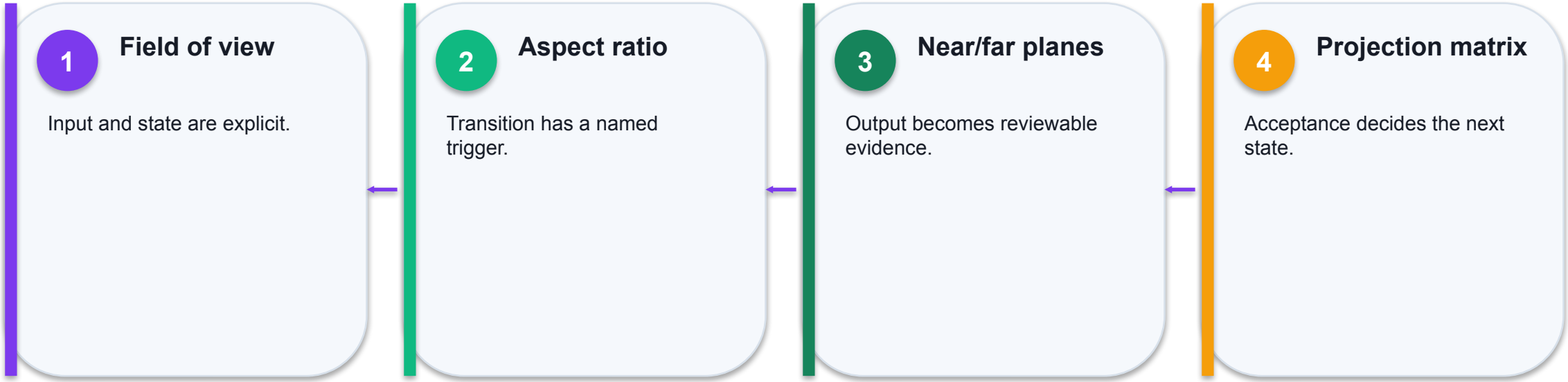
Assign placement to the root and local animation to the intended child mesh.

VERIFICATION EVIDENCE

The root follows placement while child motion preserves anchor locations.

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Perspective camera projection



MECHANISM RULE

The camera frustum maps view-space depth to clip space; tight near/far ranges improve depth precision.

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Perspective camera projection: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
camera.aspect = width / height; camera.updateProjectionMatrix();

// acceptance
A 390 px viewport preserves object proportions and no target is clipped at expected distance.
```

INPUT	Field of view
STATE	Aspect ratio
CONTROL	Near/far planes
EVIDENCE	Projection matrix

ACCEPTANCE A 390 px viewport preserves object proportions and no target is clipped at expected distance.

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TOPIC 2 · DIAGNOSTIC

Perspective camera projection: failure and control

FAILURE MODE

A resize stretches the scene or far/near values cause z-fighting and clipping.

CONTROL / RECOVERY

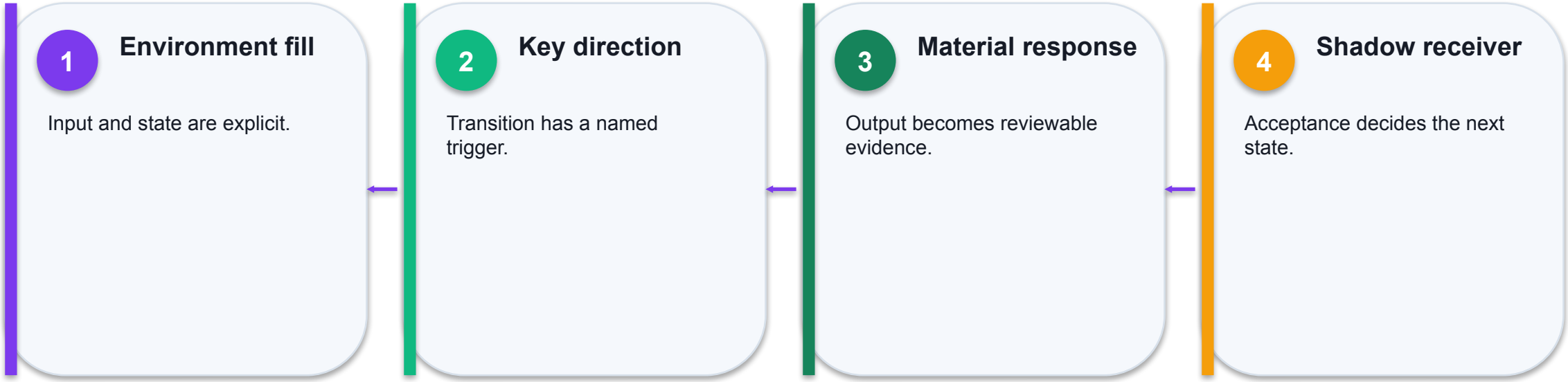
Update aspect on resize and choose the smallest practical depth range.

VERIFICATION EVIDENCE

A 390 px viewport preserves object proportions and no target is clipped at expected distance.

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Lighting model and evidence



MECHANISM RULE

Hemisphere light supplies broad fill; a directional key defines form; shadows require renderer, light, caster and receiver settings.

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Lighting model and evidence: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
renderer.shadowMap.enabled = true; key.castShadow = true;

// acceptance
The object casts a visible, correctly oriented shadow on the floor preview.
```

INPUT	Environment fill
STATE	Key direction
CONTROL	Material response
EVIDENCE	Shadow receiver

ACCEPTANCE The object casts a visible, correctly oriented shadow on the floor preview.

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Lighting model and evidence: failure and control

FAILURE MODE

Enabling only one part of the shadow chain produces no shadow while code appears plausible.

CONTROL / RECOVERY

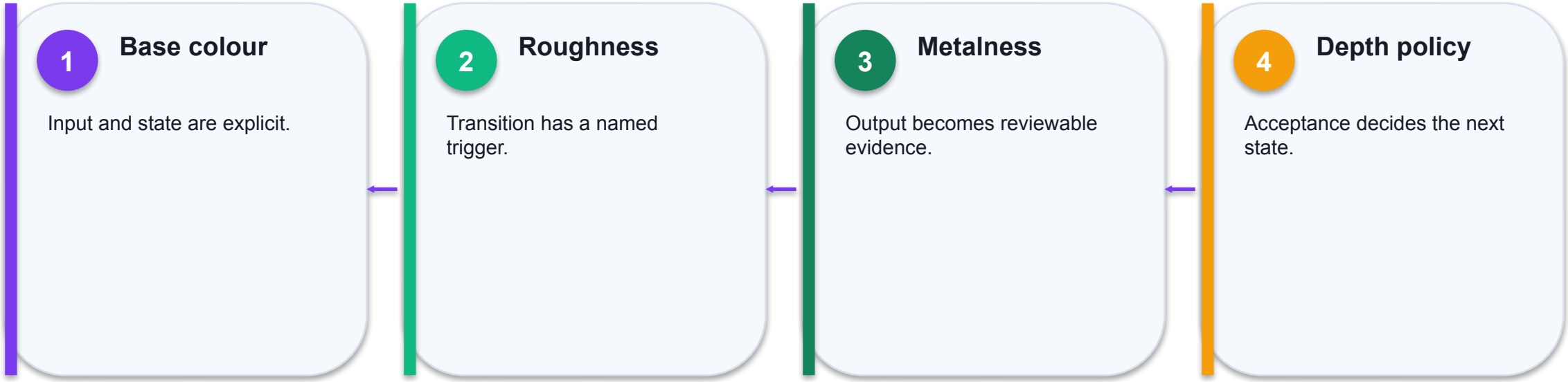
Verify the complete chain and maintain readable contrast under AR camera content.

VERIFICATION EVIDENCE

The object casts a visible, correctly oriented shadow on the floor preview.

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Material contract



MECHANISM RULE

Physically based material parameters control reflected light; depthTest/depthWrite control virtual ordering.

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Material contract: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
new THREE.MeshStandardMaterial({color, metalness:0.28, roughness:0.34})

// acceptance
Colour identity remains visible and surfaces retain form at multiple camera angles.
```

INPUT	Base colour
STATE	Roughness
CONTROL	Metalness
EVIDENCE	Depth policy

ACCEPTANCE — Colour identity remains visible and surfaces retain form at multiple camera angles.

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Material contract: failure and control

FAILURE MODE

Highly metallic content without an environment looks black or misleadingly glossy.

CONTROL / RECOVERY

Use restrained parameters and test against the actual lighting/background.

VERIFICATION EVIDENCE

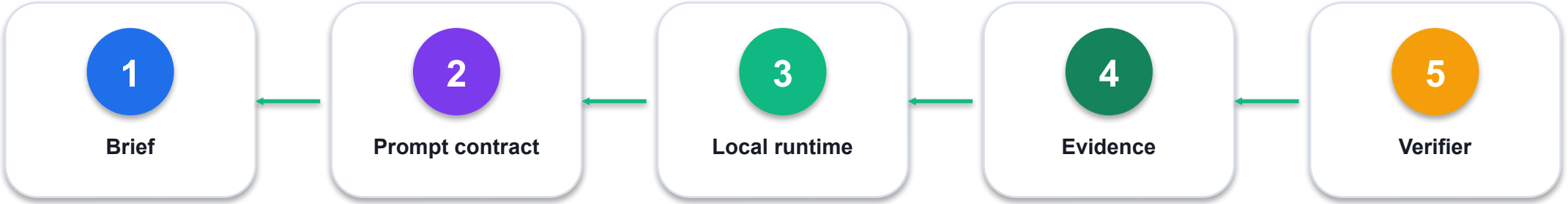
Colour identity remains visible and surfaces retain form at multiple camera angles.

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TOPIC 2 · HANDS-ON LAB · 90 MIN

Lab 04 · Scene Graph, Lighting and Animation

Build a deterministic Three.js scene graph and explain how transforms, lighting, render loop, and cleanup interact.

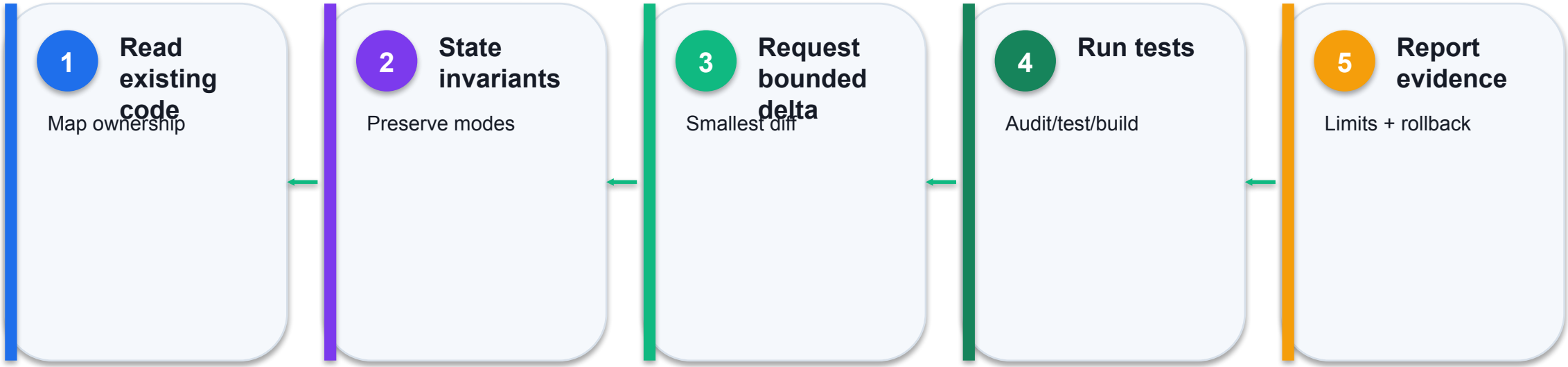


YOU'LL PRODUCE

Scene JSON · runtime evidence · review artifact · PASS Lab 04. Full detailed procedure: labs/lab-04-scene-graph-lighting-and-animation/README.md

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Lab 04: prompt-to-runtime contract



GUARDRAIL

Generated code remains a proposal until the JSON contract, runtime state and verifier agree.

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Lab 04: technical excerpt

```
{
  "targetMode": "desktop",
  "tracking": {
    "orbitCamera": true
  },
  "content": {
    "subject": "engine",
    "instances": 3,
    "callouts": 3,
    "scaleMetres": 0.65
  },
  "performance": {
    "targetFps": 60,
    "drawCallBudget": 40,
    "triangleBudget": 80000
  }
}
```

1 **ASSERT 1** generated root uses metric scale

2 **ASSERT 2** animation can be disabled

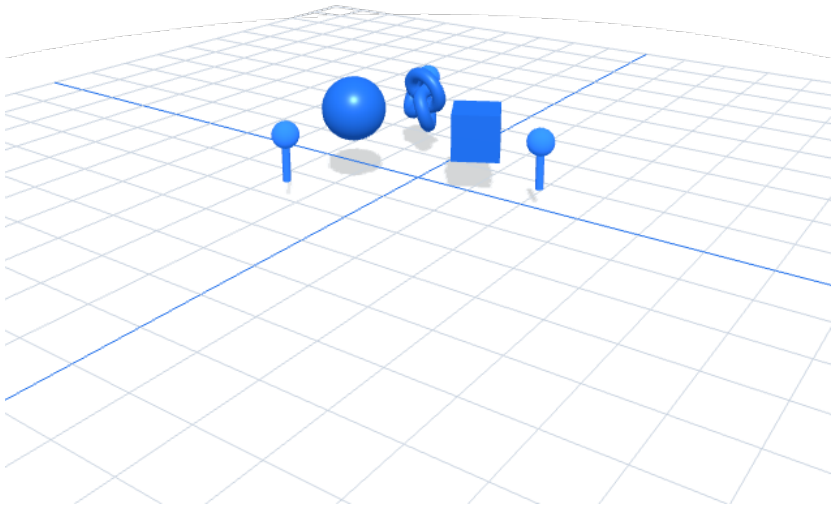
3 **ASSERT 3** performance budget is explicit

4 **ASSERT 4** cleanup/disposal is documented

RUN **node labs/lab-04-scene-graph-lighting-and-animation/verify.mjs → PASS Lab 04**

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Lab 04: acceptance evidence



PASS EVIDENCE

✓ generated root uses metric scale

✓ animation can be disabled

✓ performance budget is explicit

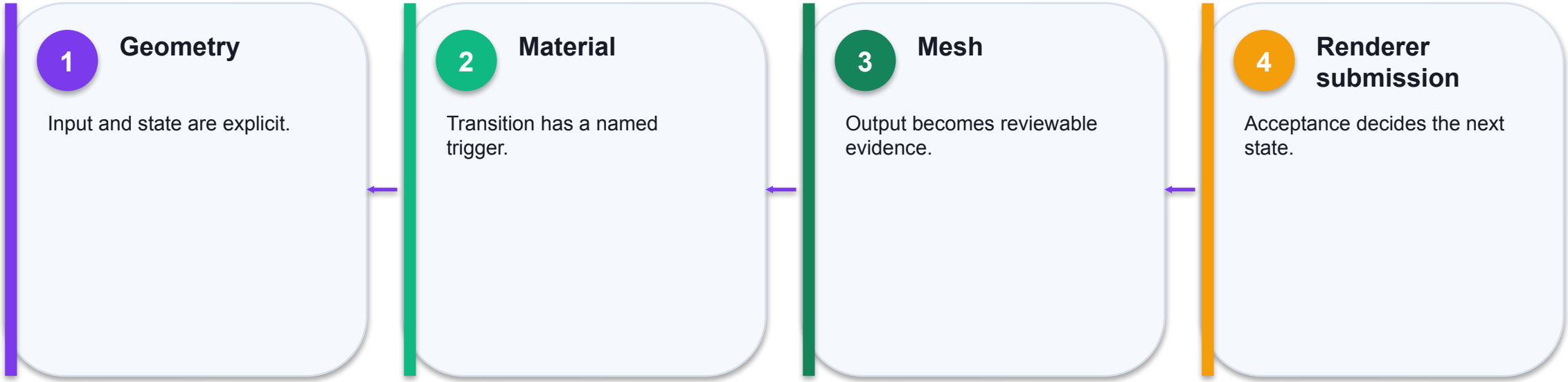
✓ cleanup/disposal is documented

DIAGNOSE

- If every child rotates around the same origin, rotate the child mesh rather than the root group.
- If shadows vanish, verify the renderer shadow map, casting light, object castShadow, and floor receiveShadow as one chain.
- If repeated generation slows down, inspect disposal of geometry and materials.

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Geometry and draw-call cost



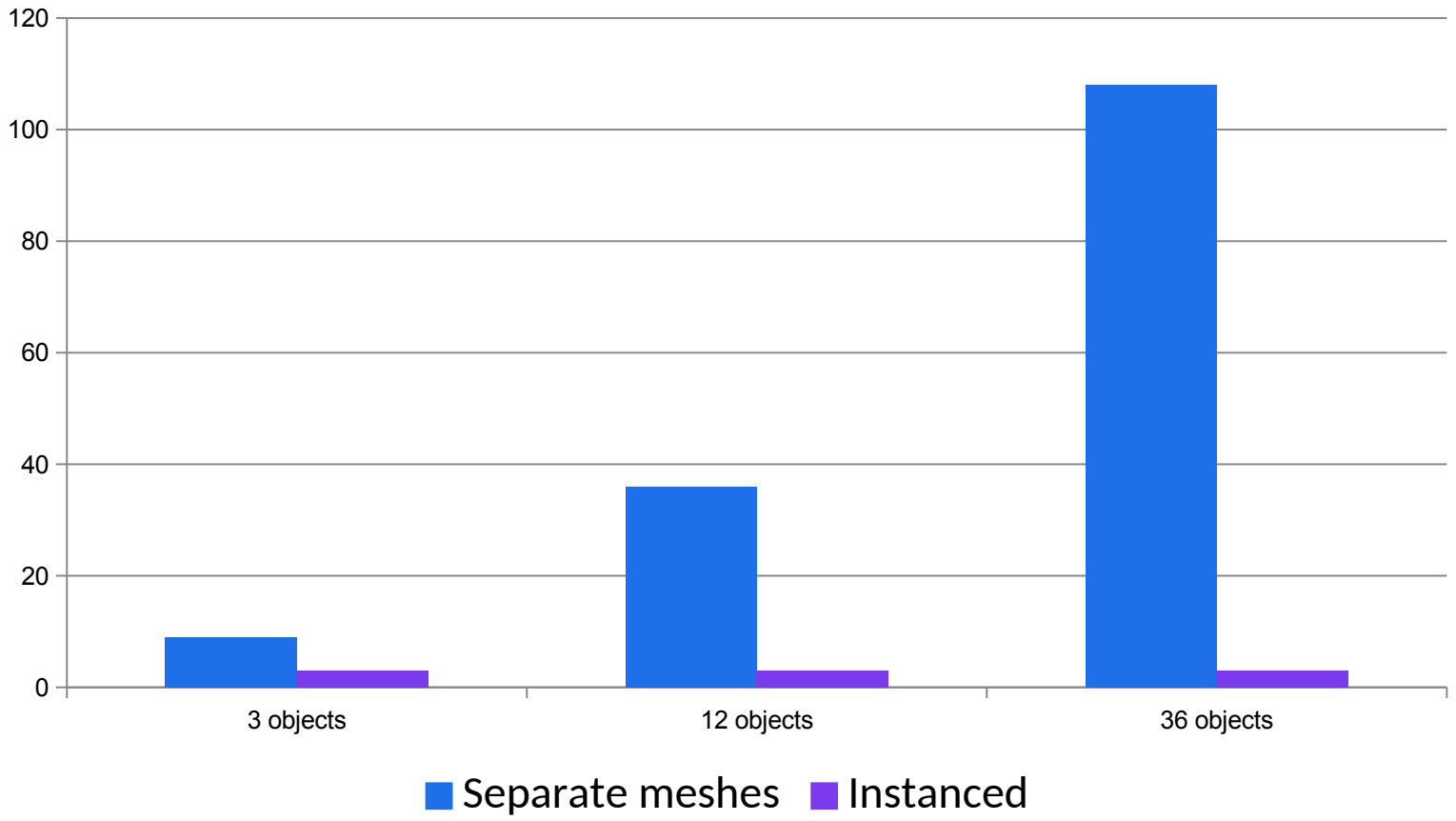
MECHANISM RULE

Each material/geometry group can create a draw call; instancing can submit repeated geometry together.

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TOPIC 2 · EDITABLE NATIVE CHART

Draw calls: separate meshes vs instancing



WHAT THE DATA SHOWS

Each material/geometry group can create a draw call; instancing can submit repeated geometry together.

DECISION RULE

Renderer stats remain within the 40-draw-call contract at the lab scene size.

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Geometry and draw-call cost: failure and control

FAILURE MODE

Dozens of identical callout meshes inflate draw calls and reduce mobile frame rate.

CONTROL / RECOVERY

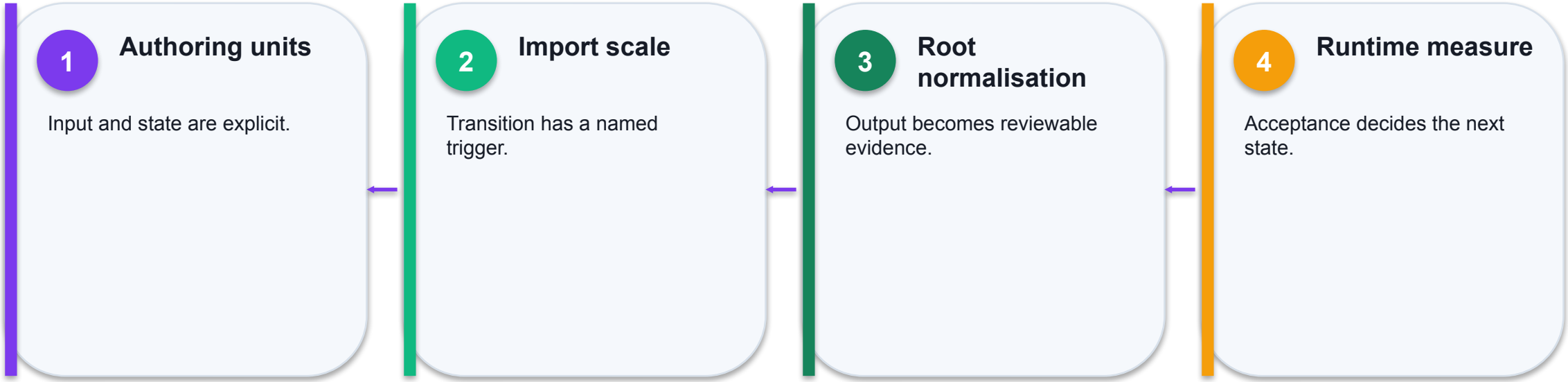
Share geometry/materials or use instancing after measuring, not by assumption.

VERIFICATION EVIDENCE

Renderer stats remain within the 40-draw-call contract at the lab scene size.

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Metric scale and asset normalisation



MECHANISM RULE

One world unit is treated as one metre; imported assets are normalised once at their boundary.

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Metric scale and asset normalisation: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
generatedRoot.scale.setScalar(contract.content.scaleMetres);

// acceptance
Bounding-box measurement matches the equipment specification within tolerance.
```

INPUT	Authoring units
STATE	Import scale
CONTROL	Root normalisation
EVIDENCE	Runtime measure

ACCEPTANCE Bounding-box measurement matches the equipment specification within tolerance.

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Metric scale and asset normalisation: failure and control

FAILURE MODE

Repeated scale corrections at parent and child levels multiply unexpectedly.

CONTROL / RECOVERY

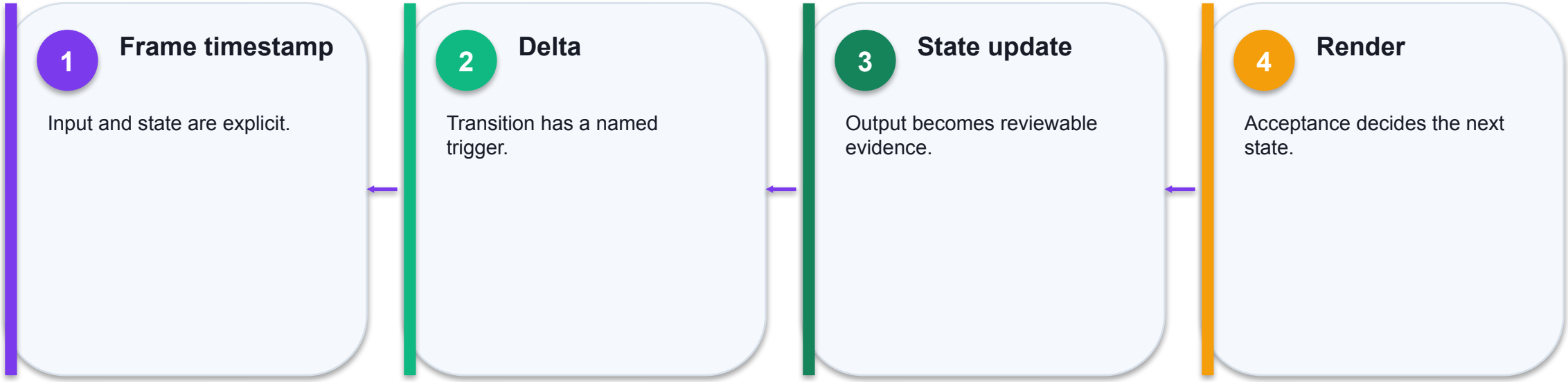
Store real size in the contract and apply exactly one normalisation factor.

VERIFICATION EVIDENCE

Bounding-box measurement matches the equipment specification within tolerance.

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Delta-time animation



MECHANISM RULE

Motion should depend on elapsed time, not frames, so behaviour remains consistent across refresh rates.

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Delta-time animation: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
rotation.y += radiansPerSecond * deltaSeconds;

// acceptance
One second produces the same angular change at 60 and 120 simulated frames.
```

INPUT	Frame timestamp
STATE	Delta
CONTROL	State update
EVIDENCE	Render

ACCEPTANCE One second produces the same angular change at 60 and 120 simulated frames.

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Delta-time animation: failure and control

FAILURE MODE

A fixed per-frame increment runs 2× faster at 120 Hz than at 60 Hz.

CONTROL / RECOVERY

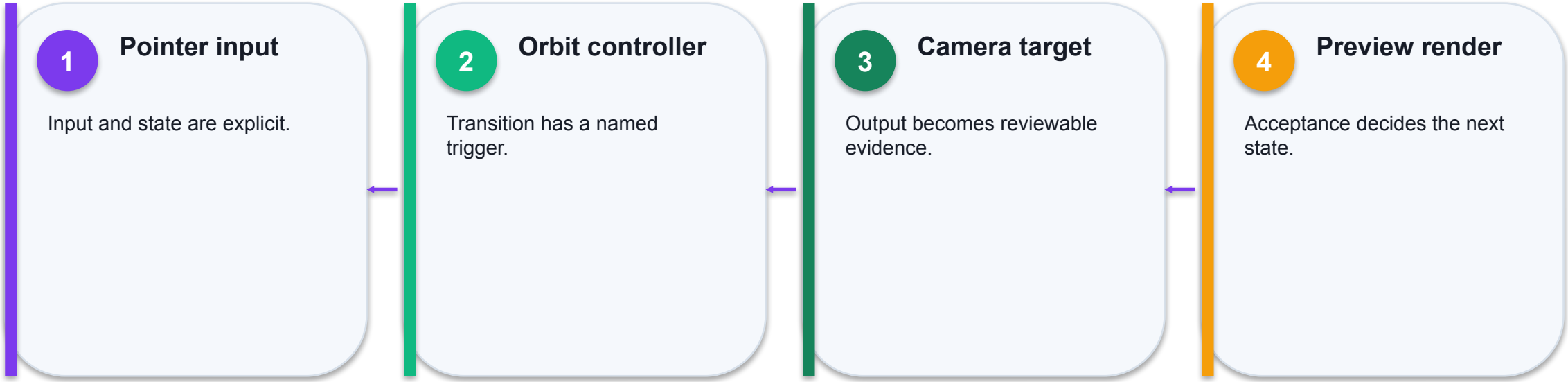
Use delta time and clamp long gaps after a suspended tab/session.

VERIFICATION EVIDENCE

One second produces the same angular change at 60 and 120 simulated frames.

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Desktop orbit fallback



MECHANISM RULE

Desktop orbit is a review mode, not simulated immersive tracking; it exposes content when WebXR is unavailable.

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Desktop orbit fallback: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
controls.enabled = !renderer.xr.isPresenting;

// acceptance
Desktop orbit works before/after XR and never overrides XR camera state.
```

INPUT	Pointer input
STATE	Orbit controller
CONTROL	Camera target
EVIDENCE	Preview render

ACCEPTANCE — Desktop orbit works before/after XR and never overrides XR camera state.

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TOPIC 2 · DIAGNOSTIC

Desktop orbit fallback: failure and control

FAILURE MODE

OrbitControls competes with XR headset camera pose during an immersive session.

CONTROL / RECOVERY

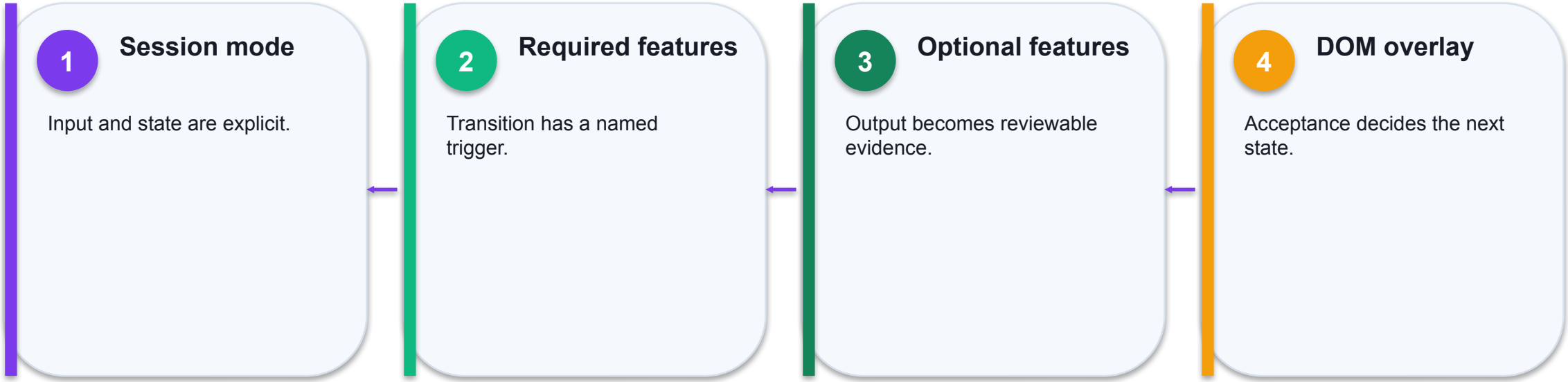
Disable desktop camera controls while XR is presenting.

VERIFICATION EVIDENCE

Desktop orbit works before/after XR and never overrides XR camera state.

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AR session feature request



MECHANISM RULE

Required features make the request fail when absent; optional features enhance the session without blocking it.

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AR session feature request: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
ARButton.createButton(renderer,{requiredFeatures:['hit-test'],optionalFeatures:['dom-overlay']})

// acceptance
The request declares hit-test as essential and desktop preview remains reachable on failure.
```

INPUT	Session mode
STATE	Required features
CONTROL	Optional features
EVIDENCE	DOM overlay

ACCEPTANCE The request declares hit-test as essential and desktop preview remains reachable on failure.

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TOPIC 2 · DIAGNOSTIC

AR session feature request: failure and control

! FAILURE MODE

Marking convenience features as required unnecessarily excludes capable devices.

✓ CONTROL / RECOVERY

Require only mechanisms essential to the lab contract and provide fallback messaging.

VERIFICATION EVIDENCE

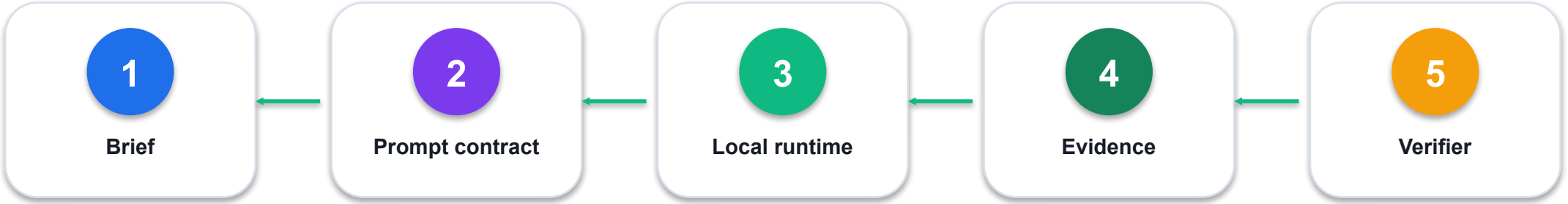
The request declares hit-test as essential and desktop preview remains reachable on failure.

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TOPIC 2 · HANDS-ON LAB · 90 MIN

Lab 05 · Plane Detection, Hit Test and Anchors

Place virtual content on a detected real-world surface using viewer-space hit testing and a visible reticle.

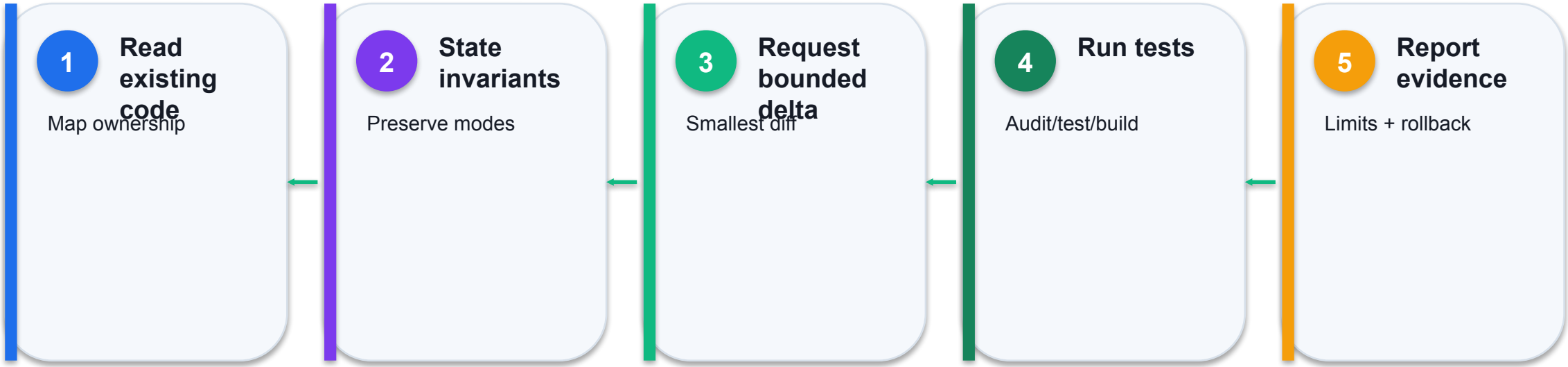


YOU'LL PRODUCE

Scene JSON · runtime evidence · review artifact · PASS Lab 05. Full detailed procedure: labs/lab-05-plane-detection-hit-test-and-anchors/README.md

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Lab 05: prompt-to-runtime contract



GUARDRAIL

Generated code remains a proposal until the JSON contract, runtime state and verifier agree.

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Lab 05: technical excerpt

```
{
  "targetMode": "ar",
  "tracking": {
    "plane": "horizontal",
    "placement": "hit-test",
    "anchorPersistence": "session"
  },
  "content": {
    "subject": "turbine",
    "instances": 3,
    "callouts": 3,
    "scaleMetres": 0.65
  },
  "performance": {
    "targetFps": 60,
    "drawCallBudget": 40,
    "triangleBudget": 80000
  }
}
```

1 **ASSERT 1** AR uses horizontal-plane hit test

2 **ASSERT 2** reticle precedes placement

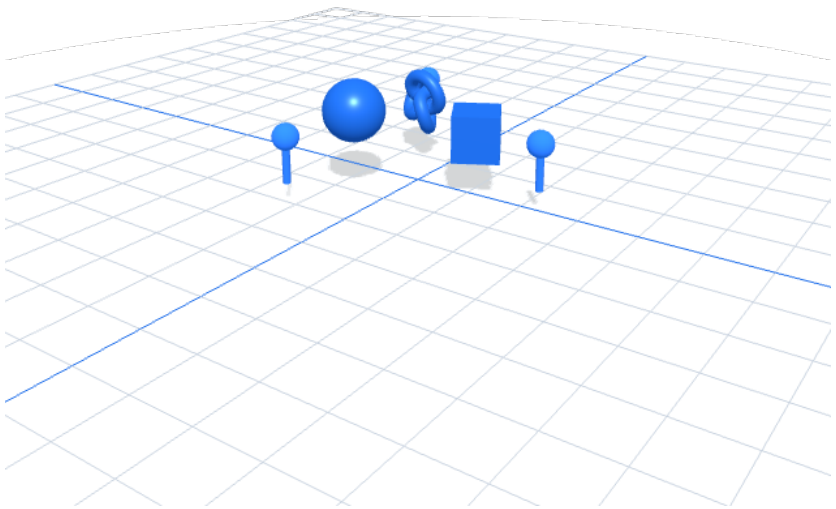
3 **ASSERT 3** session cleanup is explicit

4 **ASSERT 4** virtual object scale is metric

RUN **node labs/lab-05-plane-detection-hit-test-and-anchors/verify.mjs → PASS Lab 05**

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Lab 05: acceptance evidence



PASS EVIDENCE

✓ AR uses horizontal-plane hit test

✓ reticle precedes placement

✓ session cleanup is explicit

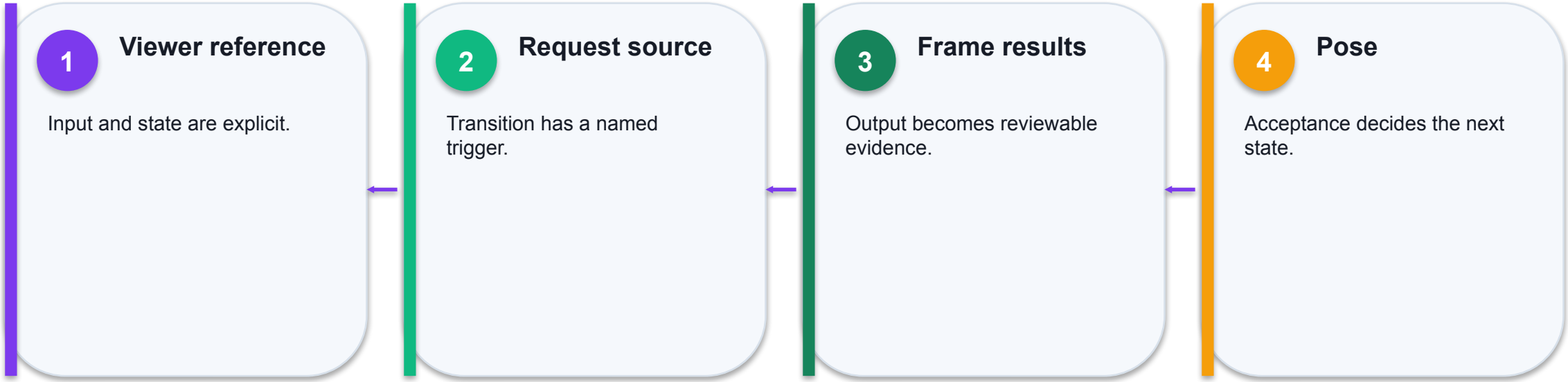
✓ virtual object scale is metric

DIAGNOSE

- No reticle: improve lighting/texture on the physical surface and move the device slowly.
- Object drifts: avoid repeatedly overwriting the placed transform after selection.
- Session will not launch: confirm WebXR AR support and localhost/HTTPS secure context.

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Viewer-space hit-test source



MECHANISM RULE

A transient hit ray originates from viewer space; results are evaluated per XRFrame against the app reference space.

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Viewer-space hit-test source: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
frame.getHitTestResults(hitTestSource)[0]?.getPose(referenceSpace)

// acceptance
Instrumentation shows one source request per start–end cycle.
```

INPUT	Viewer reference
STATE	Request source
CONTROL	Frame results
EVIDENCE	Pose

ACCEPTANCE Instrumentation shows one source request per start–end cycle.

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Viewer-space hit-test source: failure and control

FAILURE MODE

Requesting a new hit-test source every frame wastes resources and may fail.

CONTROL / RECOVERY

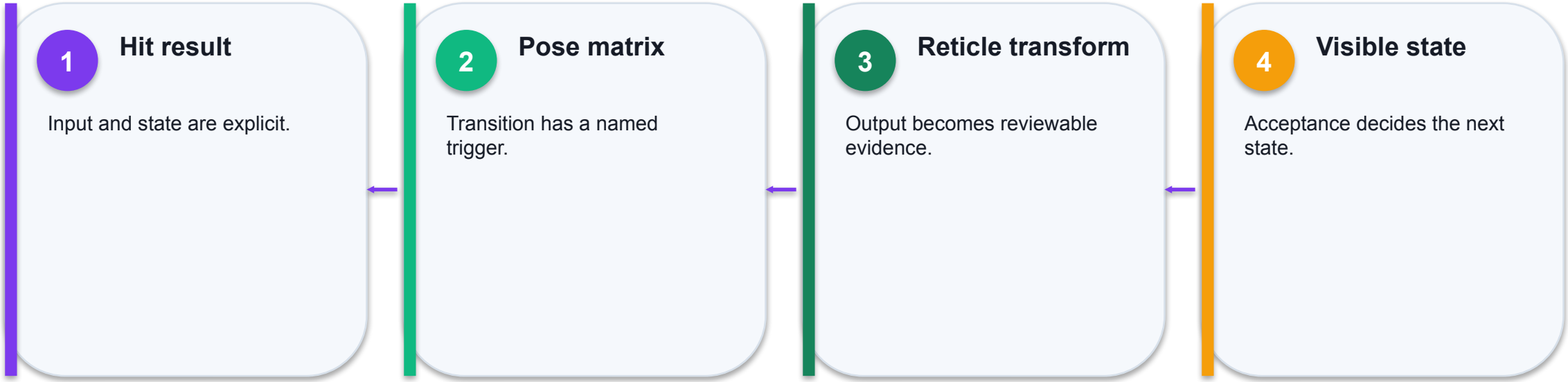
Request once per session and reuse until the session ends.

VERIFICATION EVIDENCE

Instrumentation shows one source request per start–end cycle.

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Reticle as placement evidence



MECHANISM RULE

The reticle shows the current placement pose before the learner commits content.

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Reticle as placement evidence: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
reticle.matrix.fromArray(pose.transform.matrix);

// acceptance
No result hides reticle; stable result shows it at the detected surface.
```

INPUT	Hit result
STATE	Pose matrix
CONTROL	Reticle transform
EVIDENCE	Visible state

ACCEPTANCE — **No result hides reticle; stable result shows it at the detected surface.**

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Reticle as placement evidence: failure and control

FAILURE MODE

Content appears before a valid surface or the reticle remains visible after results disappear.

CONTROL / RECOVERY

Tie reticle visibility directly to current-frame hit results.

VERIFICATION EVIDENCE

No result hides reticle; stable result shows it at the detected surface.

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Anchor and persistence boundary



MECHANISM RULE

A session-scoped placement remains stable only for the active XR session unless a persistent-anchor facility is explicitly available.

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Anchor and persistence boundary: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
anchorPersistence: 'session'

// acceptance
Reload clearly resets placement and never claims cross-session persistence.
```

INPUT	Pose selected
STATE	Root placed
CONTROL	Session tracks
EVIDENCE	Session ends

ACCEPTANCE — Reload clearly resets placement and never claims cross-session persistence.

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Anchor and persistence boundary: failure and control

FAILURE MODE

The app promises persistence across reloads without storing or resolving a durable anchor.

CONTROL / RECOVERY

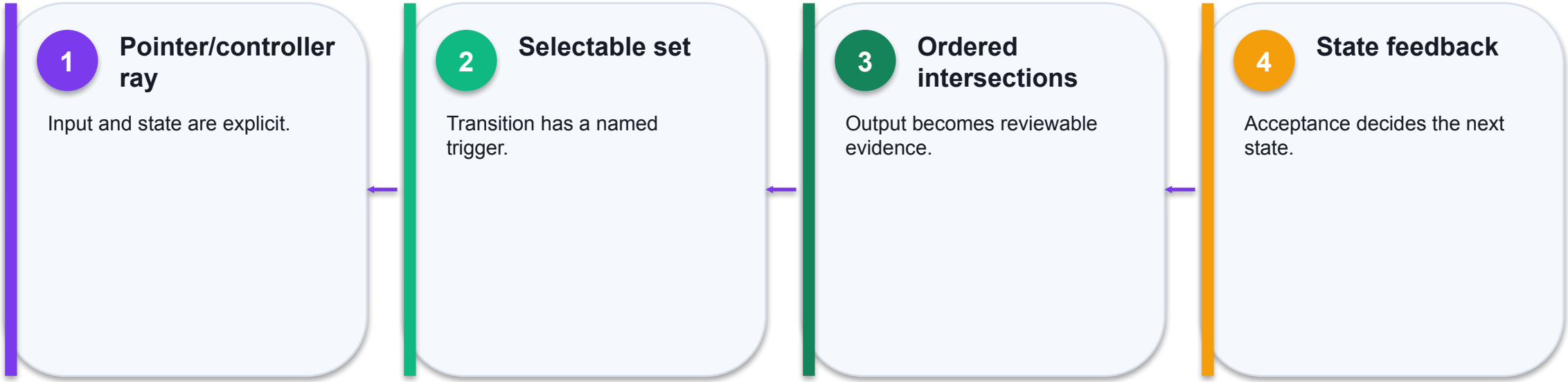
Name the persistence boundary in the scene contract and handover.

VERIFICATION EVIDENCE

Reload clearly resets placement and never claims cross-session persistence.

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Raycast selection pipeline



MECHANISM RULE

Raycasting maps an input ray to distance-sorted scene intersections.

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Raycast selection pipeline: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
const hit = raycaster.intersectObjects(selectables,true)[0] ?? null;

// acceptance
Nearest valid selectable object is highlighted; empty ray returns no-hit feedback.
```

INPUT	Pointer/controller ray
STATE	Selectable set
CONTROL	Ordered intersections
EVIDENCE	State feedback

ACCEPTANCE — Nearest valid selectable object is highlighted; empty ray returns no-hit feedback.

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TOPIC 2 · DIAGNOSTIC

Raycast selection pipeline: failure and control

FAILURE MODE

The ray tests the whole scene, selecting floor/helpers or hidden children.

CONTROL / RECOVERY

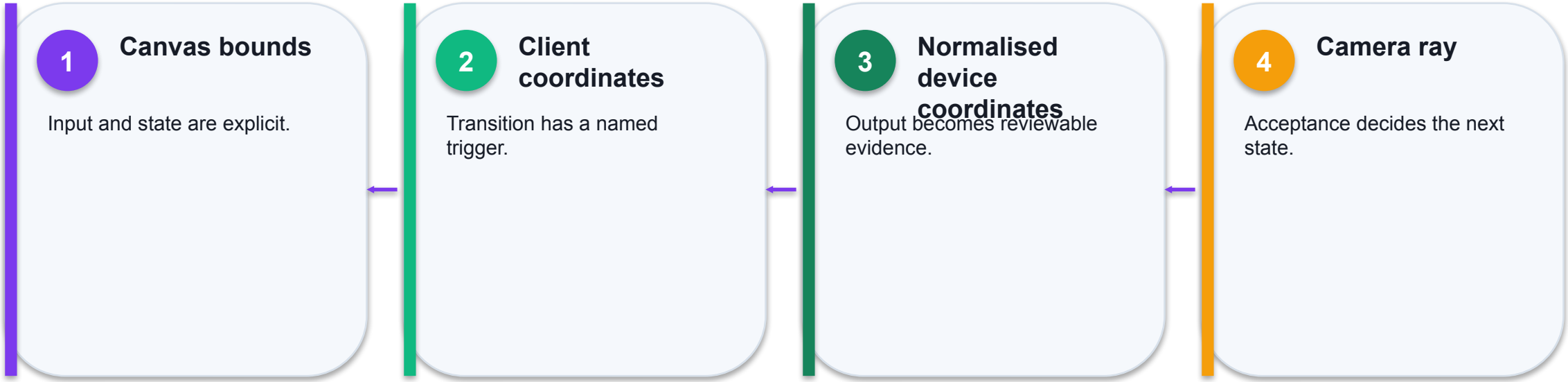
Maintain a named selectable collection and filter by state/layer.

VERIFICATION EVIDENCE

Nearest valid selectable object is highlighted; empty ray returns no-hit feedback.

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Pointer coordinate normalisation



MECHANISM RULE

Pointer x/y must be relative to the rendering canvas, then mapped to [-1,1] with inverted y.

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Pointer coordinate normalisation: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
x=((clientX-left)/width)*2-1; y=-((clientY-top)/height)*2+1;

// acceptance
Touches at all four canvas corners map close to (-1,1), (1,1), (-1,-1), (1,-1).
```

INPUT	Canvas bounds
STATE	Client coordinates
CONTROL	Normalised device coordinates
EVIDENCE	Camera ray

ACCEPTANCE Touches at all four canvas corners map close to (-1,1), (1,1), (-1,-1), (1,-1).

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Pointer coordinate normalisation: failure and control

FAILURE MODE

Using window dimensions produces mobile misses when the canvas is inset.

CONTROL / RECOVERY

Compute from the current canvas bounding rectangle for every pointer event.

VERIFICATION EVIDENCE

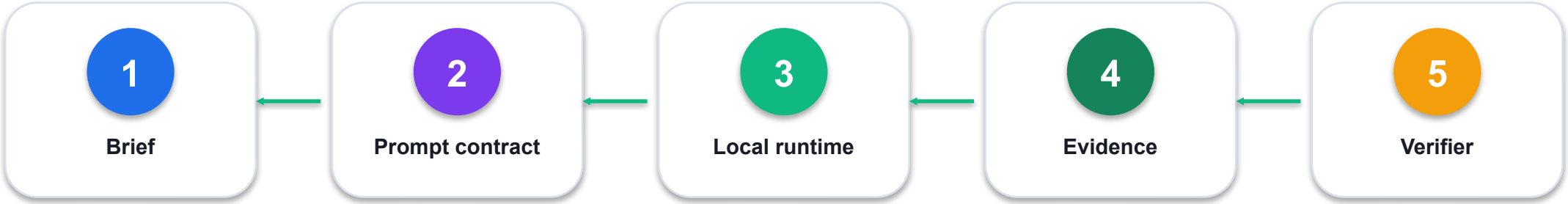
Touches at all four canvas corners map close to (-1,1), (1,1), (-1,-1), (1,-1).

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TOPIC 2 · HANDS-ON LAB · 75 MIN

Lab 06 · Raycast, Touch and Spatial Feedback

Implement selection feedback and diagnose interaction misses using ray origin, direction, target bounds, and state evidence.

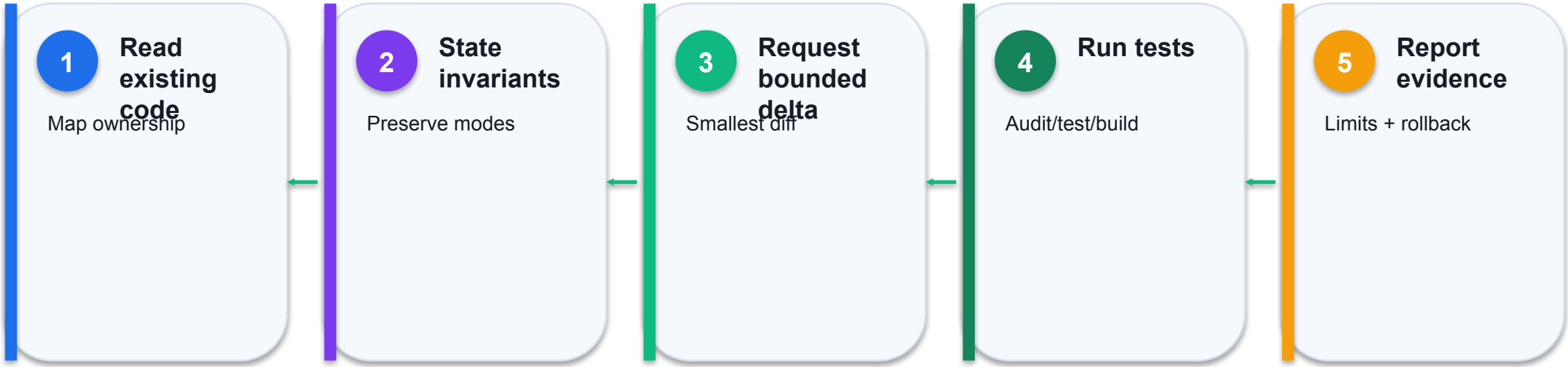


YOU'LL PRODUCE

Scene JSON · runtime evidence · review artifact · PASS Lab 06. Full detailed procedure: labs/lab-06-raycast-touch-and-spatial-feedback/README.md

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Lab 06: prompt-to-runtime contract



GUARDRAIL

Generated code remains a proposal until the JSON contract, runtime state and verifier agree.

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Lab 06: technical excerpt

```
{
  "targetMode": "ar",
  "tracking": {
    "plane": "horizontal",
    "placement": "hit-test",
    "anchorPersistence": "session"
  },
  "content": {
    "subject": "robot",
    "instances": 4,
    "callouts": 4,
    "scaleMetres": 0.65
  },
  "performance": {
    "targetFps": 60,
    "drawCallBudget": 40,
    "triangleBudget": 80000
  }
}
```

1 **ASSERT 1** tap interaction is declared

2 **ASSERT 2** nearest target is selected

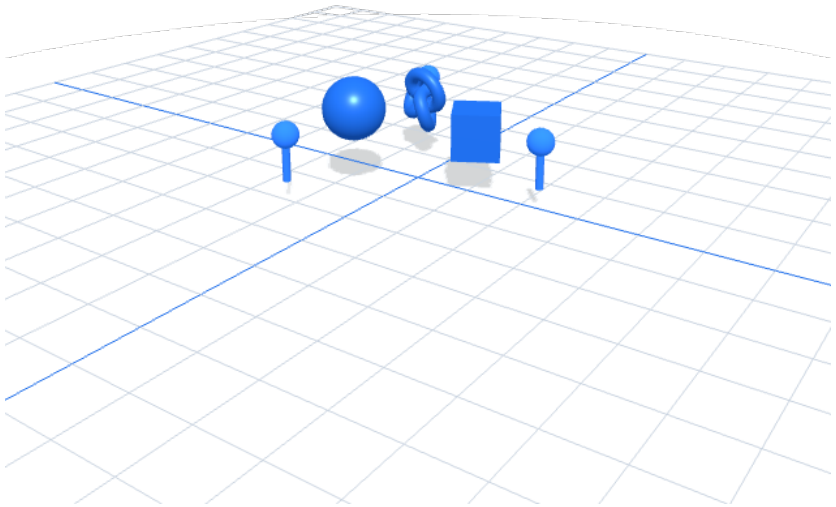
3 **ASSERT 3** no-hit state is visible

4 **ASSERT 4** interaction review identifies a cause and adjustment

RUN **node labs/lab-06-raycast-touch-and-spatial-feedback/verify.mjs → PASS Lab 06**

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Lab 06: acceptance evidence



PASS EVIDENCE

✓ tap interaction is declared

✓ nearest target is selected

✓ no-hit state is visible

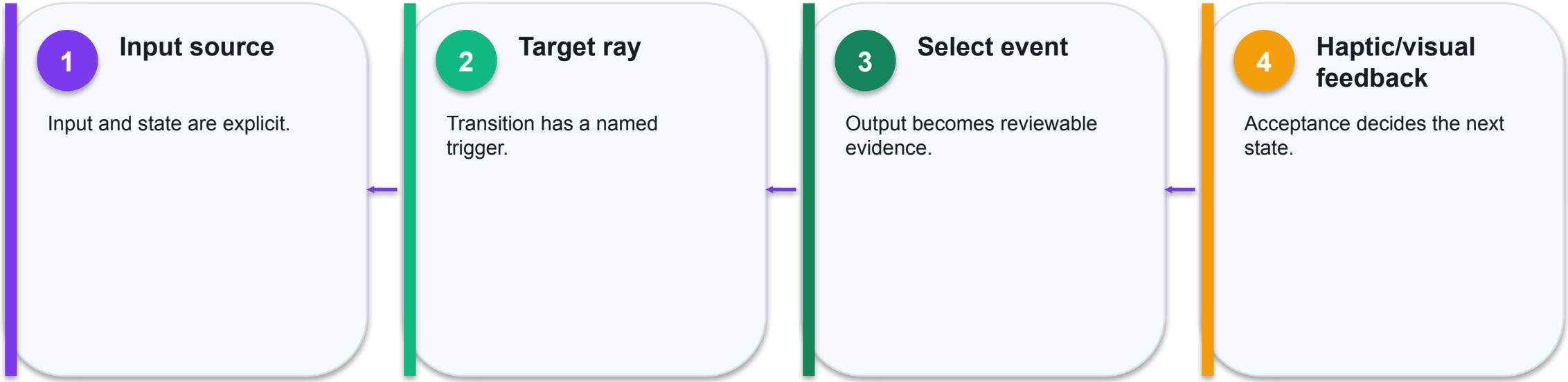
✓ interaction review identifies a cause and adjustment

DIAGNOSE

- Wrong object selected: sort intersections by distance and use the first valid selectable target.
- Touch misses on mobile: calculate pointer coordinates from the canvas bounding rectangle.
- Invisible feedback: verify material state is restored when selection changes.

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XR controller input sources



MECHANISM RULE

Tracked-pointer rays originate from the controller or hand; handedness and targetRayMode are runtime properties.

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XR controller input sources: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence

const controller = renderer.xr.getController(0);
controller.addEventListener('select', place);


// acceptance

Controller 0/1 or gaze input can trigger the same semantic action without
hard-coded handedness.
```

INPUT	Input source
STATE	Target ray
CONTROL	Select event
EVIDENCE	Haptic/visual feedback

ACCEPTANCE — **Controller 0/1 or gaze input can trigger the same semantic action without hard-coded handedness.**

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TOPIC 2 · DIAGNOSTIC

XR controller input sources: failure and control

FAILURE MODE

The implementation assumes a right-hand controller and fails for gaze or left hand.

CONTROL / RECOVERY

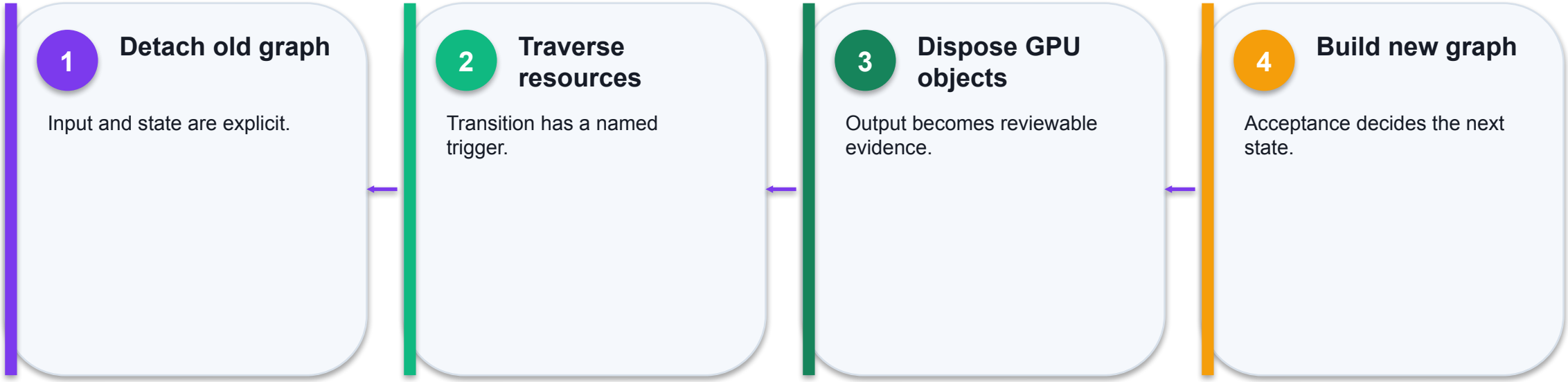
Use capability-driven input mapping and redundant feedback.

VERIFICATION EVIDENCE

Controller 0/1 or gaze input can trigger the same semantic action without hard-coded handedness.

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Scene regeneration and disposal



MECHANISM RULE

Removing a mesh does not free its GPU buffers; geometry, materials and textures need disposal when no longer used.

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Scene regeneration and disposal: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
child.traverse(o=>{o.geometry?.dispose(); o.material?.dispose();});

// acceptance
Ten regenerations show stable renderer memory counters and no console errors.
```

INPUT	Detach old graph
STATE	Traverse resources
CONTROL	Dispose GPU objects
EVIDENCE	Build new graph

ACCEPTANCE Ten regenerations show stable renderer memory counters and no console errors.

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Scene regeneration and disposal: failure and control

FAILURE MODE

Repeated prompt variations grow GPU memory and eventually reduce performance.

CONTROL / RECOVERY

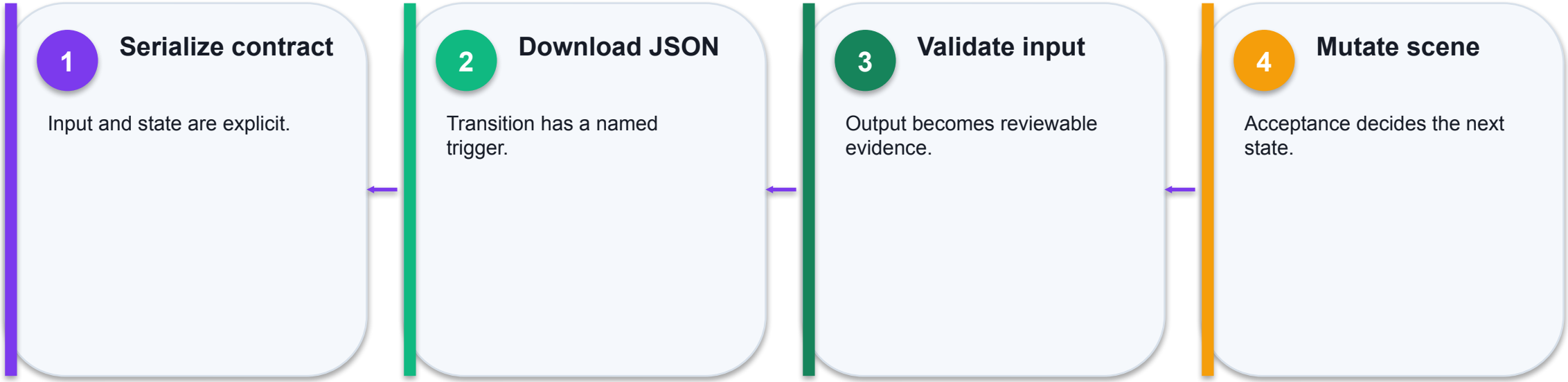
Centralise disposal at the regeneration boundary and avoid disposing shared resources prematurely.

VERIFICATION EVIDENCE

Ten regenerations show stable renderer memory counters and no console errors.

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Scene export and import boundary



MECHANISM RULE

Only validated declarative data crosses the import boundary; executable code is never accepted from scene JSON.

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Scene export and import boundary: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
const scene = safeImport(await file.text());

// acceptance
Valid export round-trips; invalid schema leaves the previous scene unchanged.
```

INPUT	Serialize contract
STATE	Download JSON
CONTROL	Validate input
EVIDENCE	Mutate scene

ACCEPTANCE Valid export round-trips; invalid schema leaves the previous scene unchanged.

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Scene export and import boundary: failure and control

! FAILURE MODE

An imported object contains unexpected code-like values or missing fields and partially mutates state.

✓ CONTROL / RECOVERY

Parse, validate schema/checks, then apply atomically.

VERIFICATION EVIDENCE

Valid export round-trips; invalid schema leaves the previous scene unchanged.

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03

TOPIC 3 · K7 · A3 · A4

Testing, Reviewing and Optimising AR/VR Applications

AR/VR boundary · comfort · frame budgets · diagnostics · handover

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MECHANISM · ANNOTATED VISUAL

VR interaction and comfort boundaries



1 **Reference space**

Local-floor origin and play-area assumptions define the coordinate contract.

2 **Controller rays**

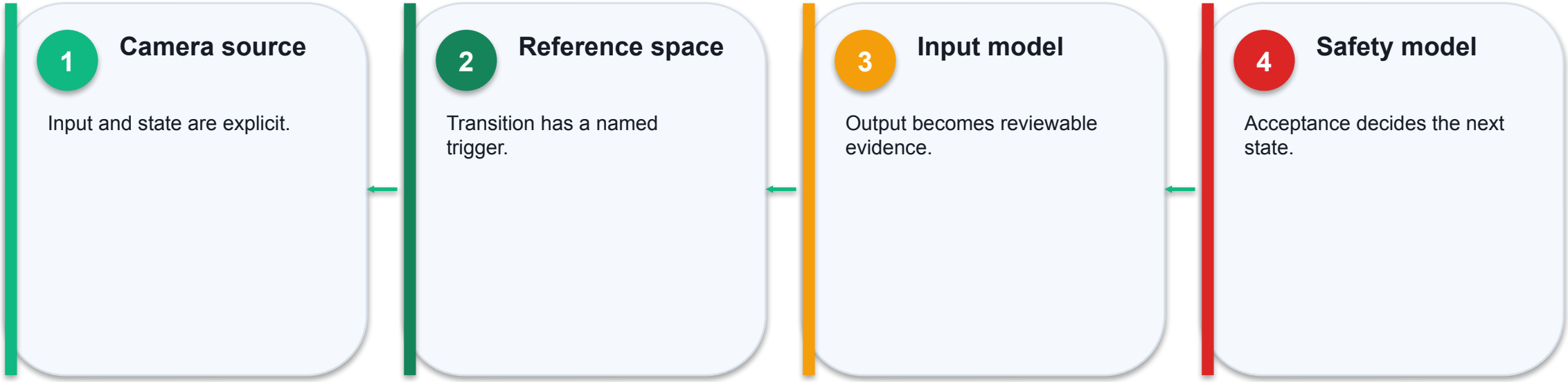
Tracked input maps to selectable bounds and visible feedback.

3 **Frame budget**

Refresh rate sets the time available for update and stereo render.

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AR and VR runtime boundary



MECHANISM RULE

AR composites virtual content with the real environment; VR replaces the view and introduces play-area/comfort constraints.

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AR and VR runtime boundary: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
targetMode: 'ar' | 'vr'

// acceptance
A reviewer can name five concrete fields that change between AR and VR.
```

INPUT	Camera source
STATE	Reference space
CONTROL	Input model
EVIDENCE	Safety model

ACCEPTANCE — **A reviewer can name five concrete fields that change between AR and VR.**

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TOPIC 3 · DIAGNOSTIC

AR and VR runtime boundary: failure and control

FAILURE MODE

A use-case comparison hides the implementation differences and reuses incompatible assumptions.

CONTROL / RECOVERY

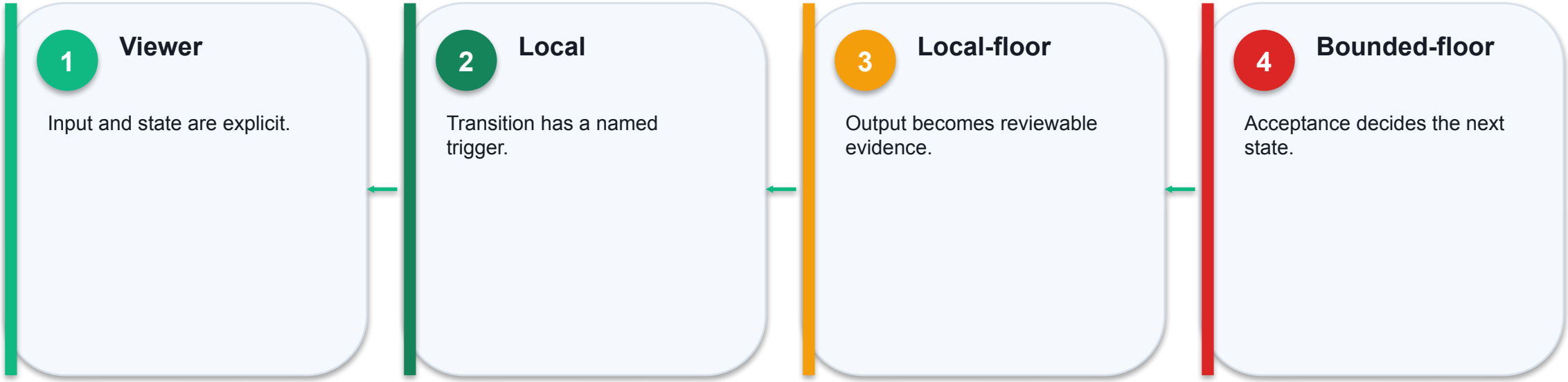
Make the decision from environment, tracking, input, safety and performance contracts.

VERIFICATION EVIDENCE

A reviewer can name five concrete fields that change between AR and VR.

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Reference-space choice



MECHANISM RULE

Reference spaces express different origin and movement assumptions; local-floor supplies floor-relative height where supported.

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Reference-space choice: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
renderer.xr.setReferenceSpaceType('local-floor');

// acceptance
Content origin and eye height match the selected reference-space contract.
```

INPUT	Viewer
STATE	Local
CONTROL	Local-floor
EVIDENCE	Bounded-floor

ACCEPTANCE — **Content origin and eye height match the selected reference-space contract.**

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Reference-space choice: failure and control

FAILURE MODE

A seated user or uncalibrated floor gets an incorrect content height.

CONTROL / RECOVERY

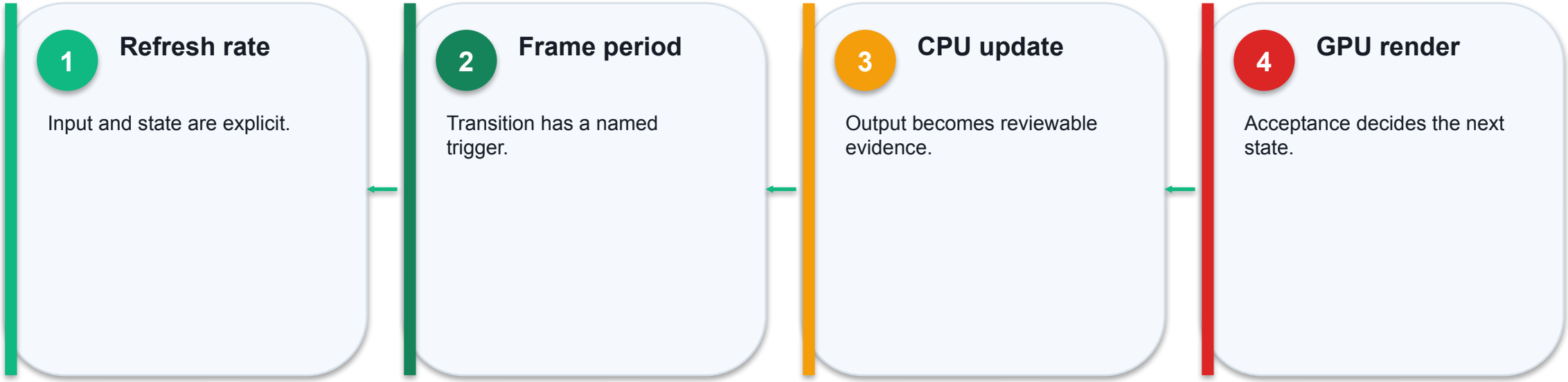
Provide seated fallback and calibrate/bound the expected play space.

VERIFICATION EVIDENCE

Content origin and eye height match the selected reference-space contract.

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Stereo frame budget

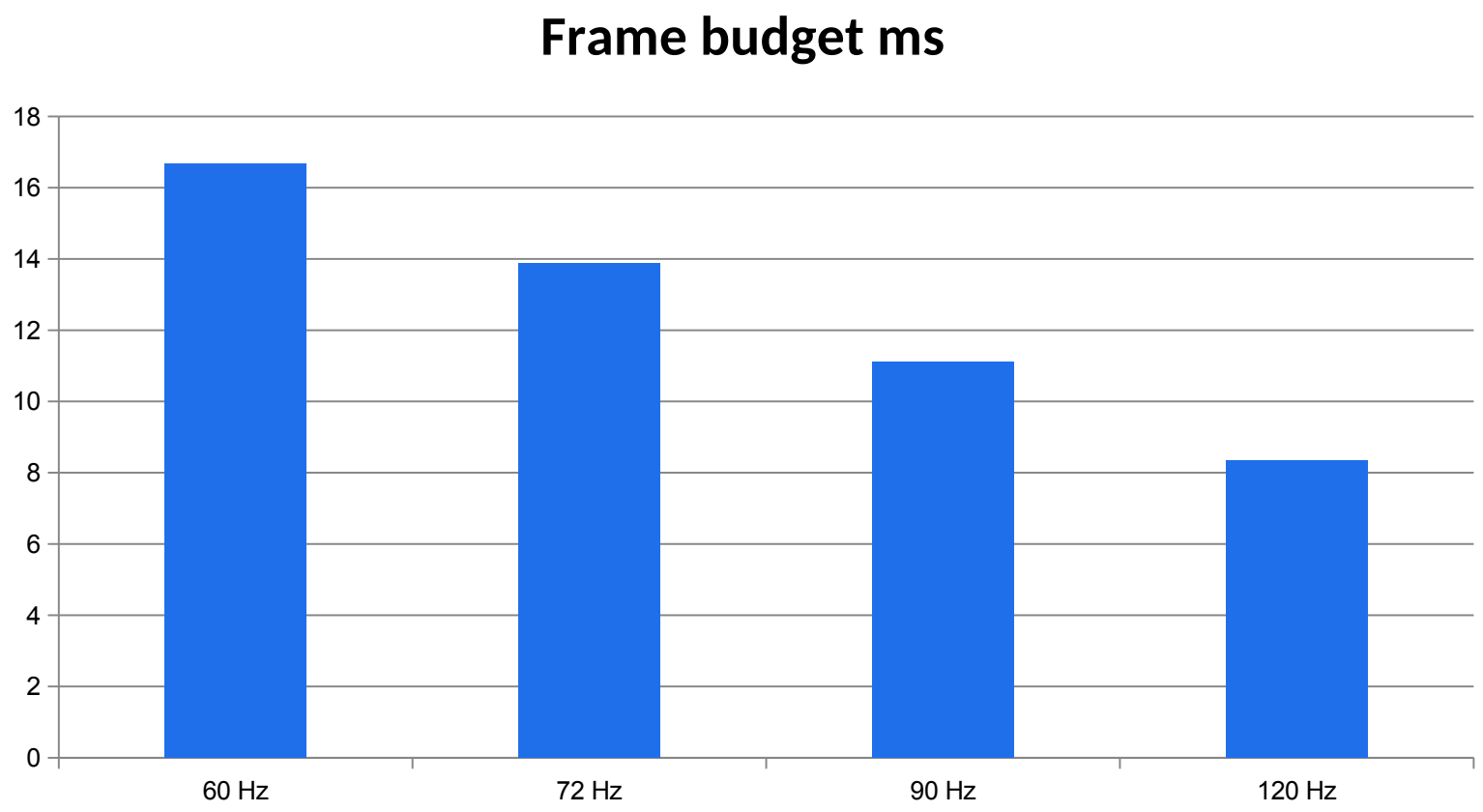


MECHANISM RULE

Frame budget in milliseconds is 1000 ÷ refresh rate: 16.7 ms at 60 Hz, 13.9 ms at 72 Hz, 11.1 ms at 90 Hz.

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Refresh rate determines the frame-time budget



WHAT THE DATA SHOWS

Frame budget in milliseconds is 1000 ÷ refresh rate: 16.7 ms at 60 Hz, 13.9 ms at 72 Hz, 11.1 ms at 90 Hz.

DECISION RULE

Measured total stays below the selected refresh-period budget.

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Stereo frame budget: failure and control

FAILURE MODE

A 16 ms frame passes desktop 60 Hz but misses a 90 Hz headset deadline.

CONTROL / RECOVERY

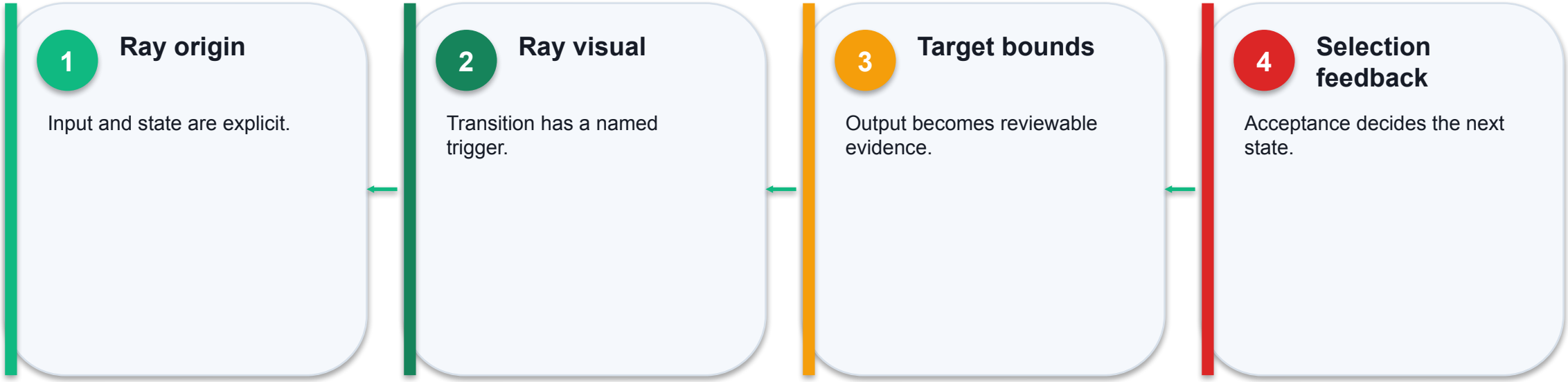
Set the target from the actual device/session and measure CPU/GPU time separately.

VERIFICATION EVIDENCE

Measured total stays below the selected refresh-period budget.

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Controller rays and affordance



MECHANISM RULE

A visible ray communicates reach and direction; targets need sufficient angular size and depth separation.

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Controller rays and affordance: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
controller.add(new THREE.Line(rayGeometry, rayMaterial));

// acceptance
Learner can predict the selected target before pressing the controller trigger.
```

INPUT	Ray origin
STATE	Ray visual
CONTROL	Target bounds
EVIDENCE	Selection feedback

ACCEPTANCE — Learner can predict the selected target before pressing the controller trigger.

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Controller rays and affordance: failure and control

! FAILURE MODE

An invisible ray makes controller selection feel random even when code works.

✓ CONTROL / RECOVERY

Show ray/reticle feedback and enlarge hit bounds without misrepresenting the visible object.

VERIFICATION EVIDENCE

Learner can predict the selected target before pressing the controller trigger.

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Teleport locomotion



MECHANISM RULE

Teleport moves the player rig to a valid floor point while maintaining head-relative height.

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Teleport locomotion: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
if (hit.surface === 'walkable') rig.position.copy(hit.point);

// acceptance
Invalid surfaces never accept; valid target shows preview and deterministic landing.
```

INPUT	Aim parabola/ray
STATE	Validate floor
CONTROL	Preview target
EVIDENCE	Commit pose

ACCEPTANCE Invalid surfaces never accept; valid target shows preview and deterministic landing.

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Teleport locomotion: failure and control

FAILURE MODE

Teleport lands inside geometry or continuous motion causes discomfort.

CONTROL / RECOVERY

Whitelist walkable surfaces, show destination, then commit on release.

VERIFICATION EVIDENCE

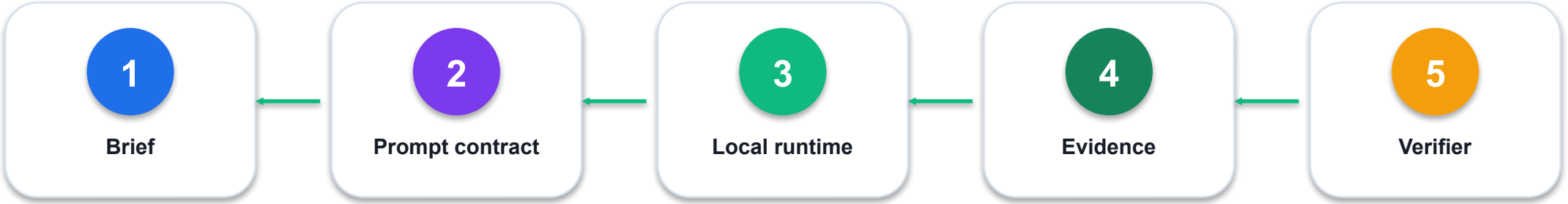
Invalid surfaces never accept; valid target shows preview and deterministic landing.

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TOPIC 3 · HANDS-ON LAB · 75 MIN

Lab 07 · Immersive VR, Controllers and Comfort

Convert the same content contract into a VR runtime and distinguish AR overlay constraints from VR room-scale constraints.

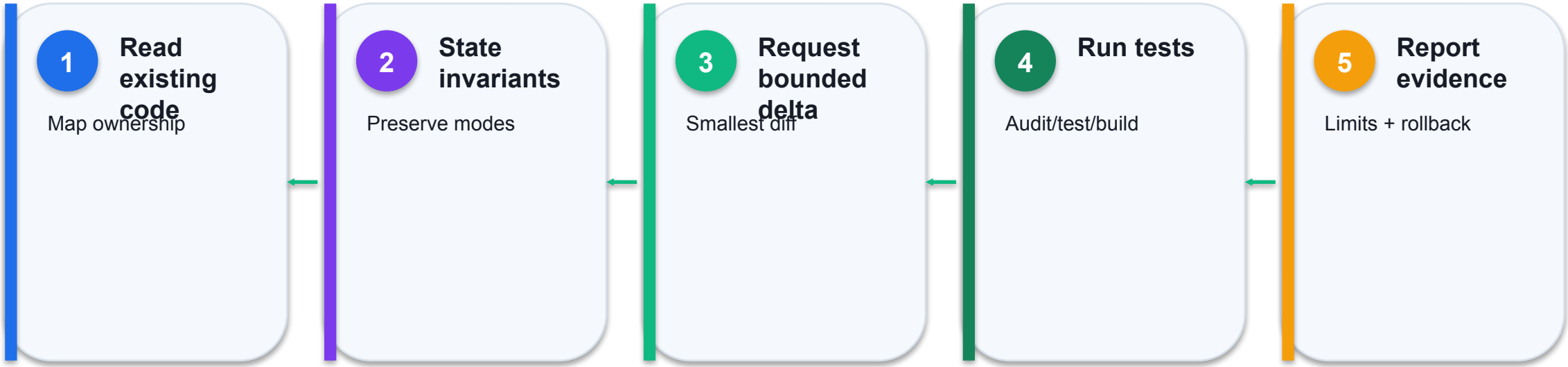


YOU'LL PRODUCE

Scene JSON · runtime evidence · review artifact · PASS Lab 07. Full detailed procedure: labs/lab-07-immersive-vr-controllers-and-comfort/README.md

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Lab 07: prompt-to-runtime contract



GUARDRAIL

Generated code remains a proposal until the JSON contract, runtime state and verifier agree.

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Lab 07: technical excerpt

```
{
  "targetMode": "vr",
  "tracking": {
    "origin": "local-floor",
    "guardian": true
  },
  "content": {
    "subject": "machine",
    "instances": 4,
    "callouts": 4,
    "scaleMetres": 0.65
  },
  "performance": {
    "targetFps": 72,
    "drawCallBudget": 40,
    "triangleBudget": 80000
  }
}
```

1 **ASSERT 1** target mode is VR

2 **ASSERT 2** local-floor origin is declared

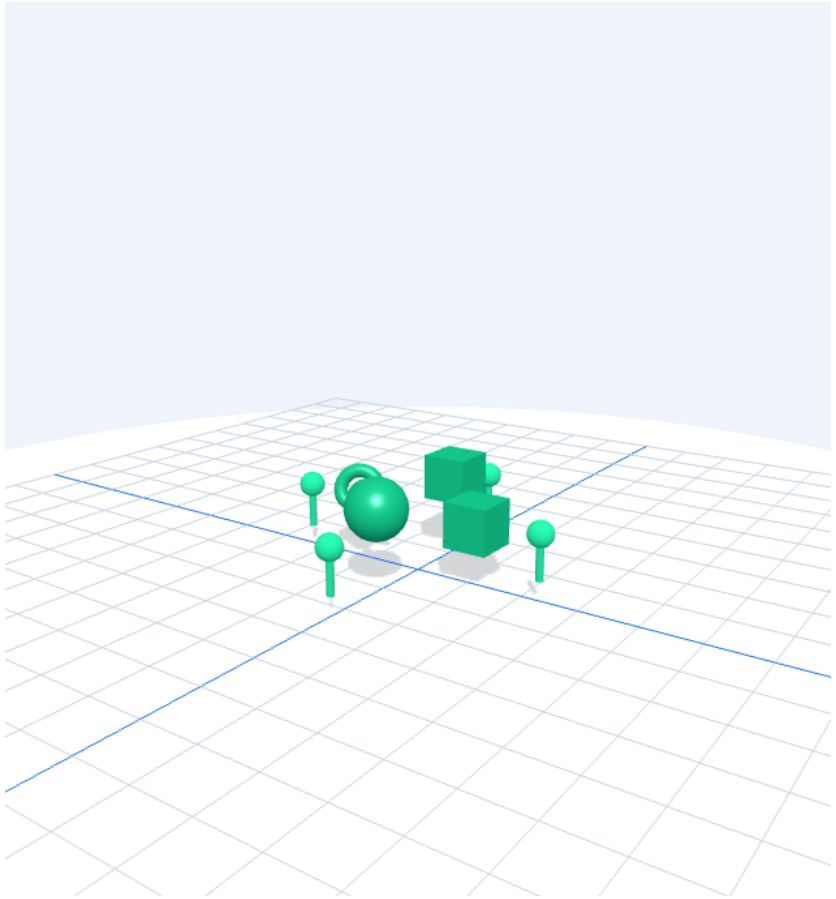
3 **ASSERT 3** guardian and seated fallback are present

4 **ASSERT 4** target frame rate is at least 72 FPS

RUN **node labs/lab-07-immersive-vr-controllers-and-comfort/verify.mjs → PASS Lab 07**

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Lab 07: acceptance evidence



PASS EVIDENCE

✓ target mode is VR

✓ local-floor origin is declared

✓ guardian and seated fallback are present

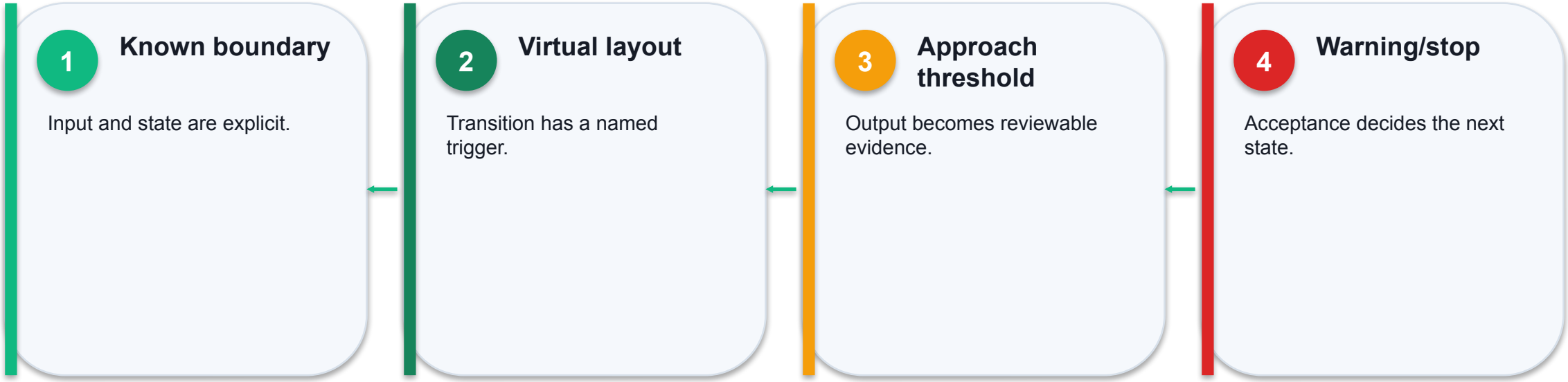
✓ target frame rate is at least 72 FPS

DIAGNOSE

- VR button disabled: verify `immersive-vr` support and the secure context.
- Floor height wrong: use local-floor reference space and calibrate the headset boundary.
- Discomfort: disable continuous motion and use teleport plus snap turns.

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Guardian and physical safety



MECHANISM RULE

VR content must respect the platform play-area boundary; applications should avoid encouraging rapid movement near edges.

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Guardian and physical safety: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
guardian: true, seatedFallback: true

// acceptance
All required targets remain reachable without crossing the physical boundary.
```

INPUT	Known boundary
STATE	Virtual layout
CONTROL	Approach threshold
EVIDENCE	Warning/stop

ACCEPTANCE All required targets remain reachable without crossing the physical boundary.

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Guardian and physical safety: failure and control

!

FAILURE MODE

A required reach exceeds the play area or training ignores real obstacles.

✓

CONTROL / RECOVERY

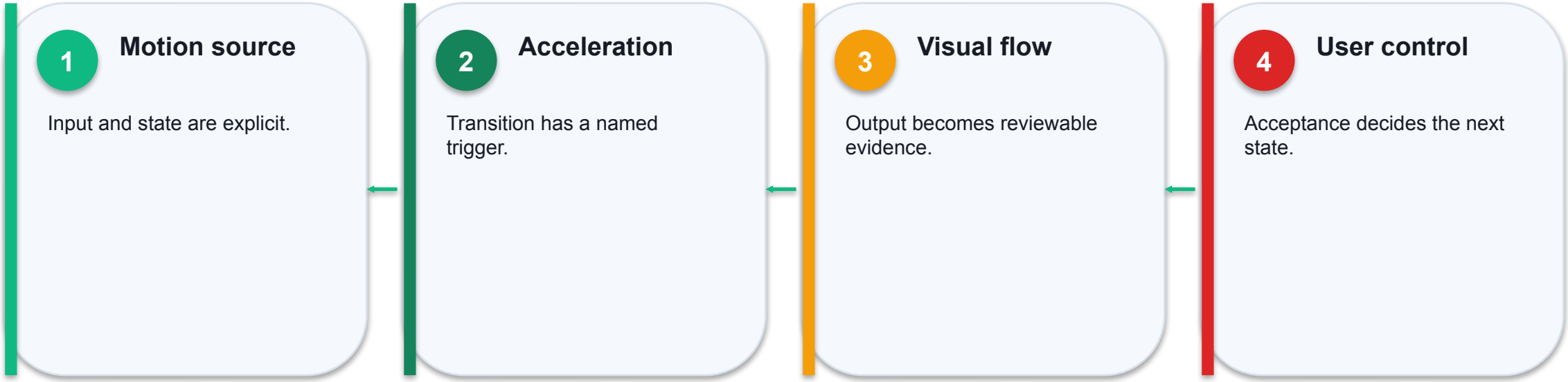
Design seated/standing variants and keep interactions within conservative reach.

VERIFICATION EVIDENCE

All required targets remain reachable without crossing the physical boundary.

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Motion comfort controls



MECHANISM RULE

Artificial camera motion creates optic flow without matching vestibular input; teleport, snap turns and reduced motion reduce conflict.

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Motion comfort controls: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
reducedMotionToggle: true

// acceptance
Reduced-motion setting stops non-essential animation immediately.
```

INPUT	Motion source
STATE	Acceleration
CONTROL	Visual flow
EVIDENCE	User control

ACCEPTANCE Reduced-motion setting stops non-essential animation immediately.

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Motion comfort controls: failure and control

! FAILURE MODE

Auto-rotation or smooth locomotion cannot be disabled.

✓ CONTROL / RECOVERY

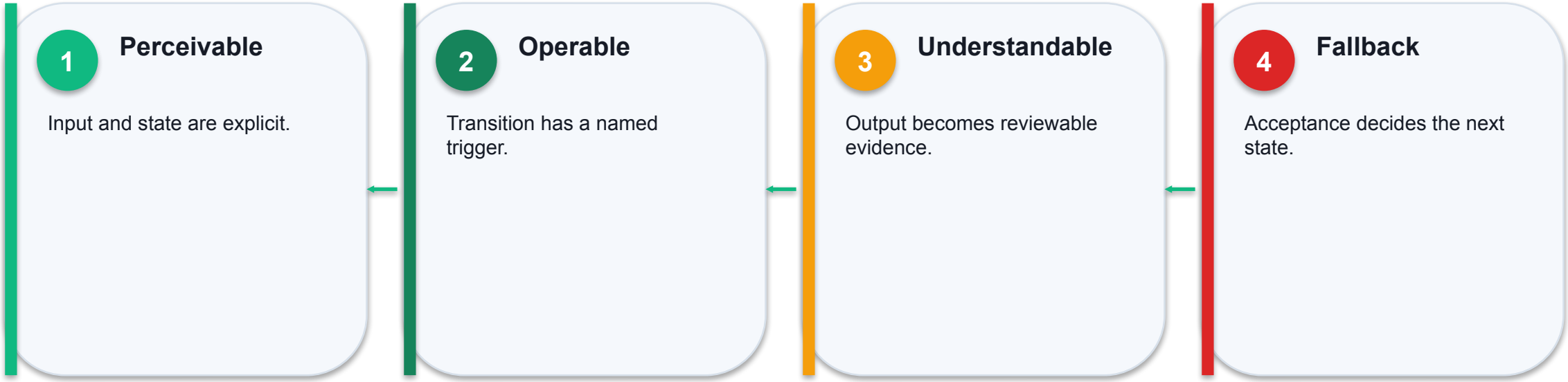
Expose motion control and default to the least provocative movement compatible with the task.

VERIFICATION EVIDENCE

Reduced-motion setting stops non-essential animation immediately.

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Spatial accessibility



MECHANISM RULE

Use redundant colour/shape/text cues, reachable controls, readable labels and a non-immersive route.

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Spatial accessibility: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
highContrastLabels: true, seatedFallback: true

// acceptance
Core task can be completed from desktop and status is not colour-only.
```

INPUT	Perceivable
STATE	Operable
CONTROL	Understandable
EVIDENCE	Fallback

ACCEPTANCE — Core task can be completed from desktop and status is not colour-only.

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Spatial accessibility: failure and control

FAILURE MODE

Meaning is encoded only by colour or requires a controller gesture unavailable to some users.

CONTROL / RECOVERY

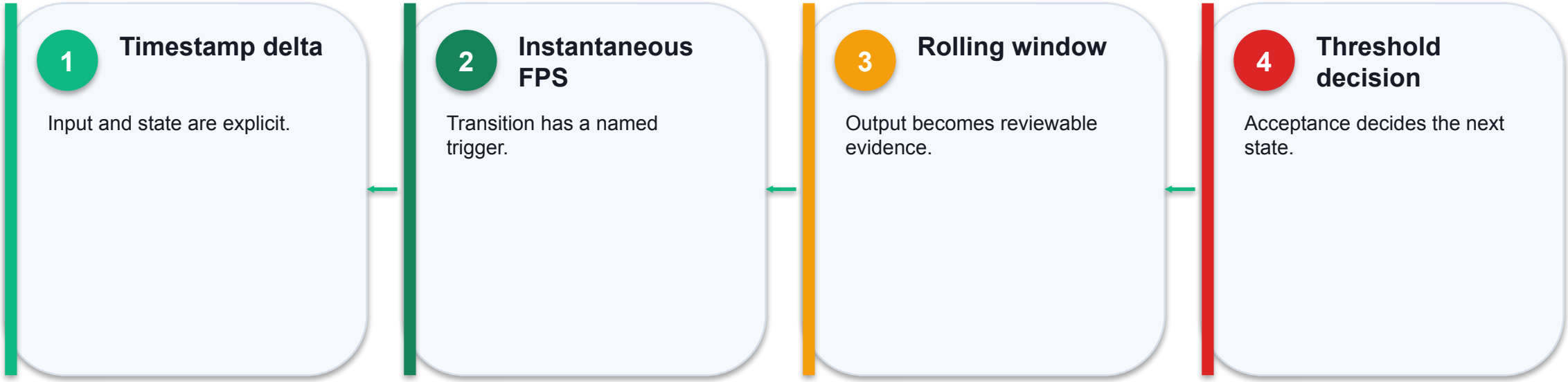
Provide multiple feedback channels and keyboard/pointer desktop access.

VERIFICATION EVIDENCE

Core task can be completed from desktop and status is not colour-only.

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FPS measurement

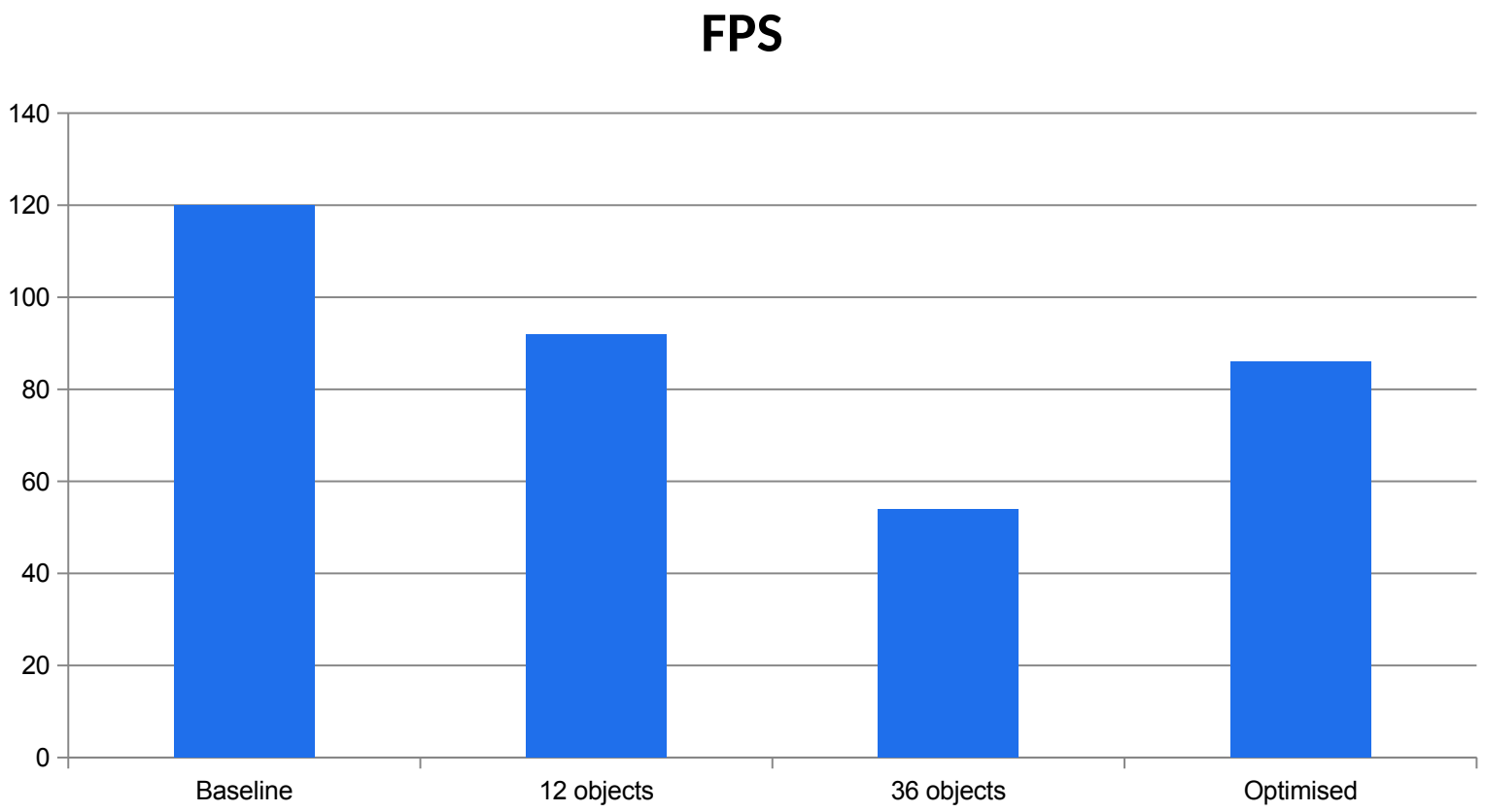


MECHANISM RULE

FPS = 1000 ÷ frame delta ms; a rolling mean reduces jitter but can hide spikes, so retain percentile/spike evidence when tuning.

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Measured scene load and optimisation



WHAT THE DATA SHOWS

FPS = 1000 ÷ frame delta ms; a rolling mean reduces jitter but can hide spikes, so retain percentile/spike evidence when tuning.

DECISION RULE

Rolling FPS meets target and no sustained frame exceeds the relevant budget.

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FPS measurement: failure and control

!

FAILURE MODE

A single 120 FPS reading is reported as stable performance.

✓

CONTROL / RECOVERY

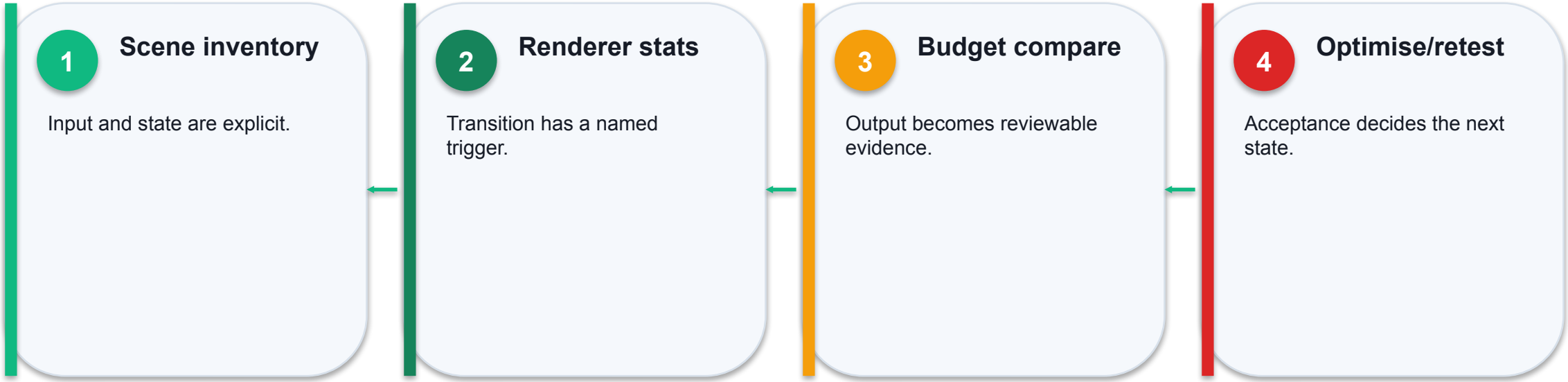
Use a representative window and record device, scene and mode.

VERIFICATION EVIDENCE

Rolling FPS meets target and no sustained frame exceeds the relevant budget.

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Draw-call and triangle budgets



MECHANISM RULE

Triangles affect vertex/fragment work; draw calls add submission/state overhead. Both must be measured.

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Draw-call and triangle budgets: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
renderer.info.render.calls; renderer.info.render.triangles;

// acceptance
Calls ≤ 40 and triangles ≤ 80,000 for the lab contract; FPS returns above target.
```

INPUT	Scene inventory
STATE	Renderer stats
CONTROL	Budget compare
EVIDENCE	Optimise/retest

ACCEPTANCE — **Calls ≤ 40 and triangles ≤ 80,000 for the lab contract; FPS returns above target.**

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TOPIC 3 · DIAGNOSTIC

Draw-call and triangle budgets: failure and control

! FAILURE MODE

The team lowers polygon count while draw-call overhead remains the actual bottleneck.

✓ CONTROL / RECOVERY

Change one budget driver at a time and retest the same scene.

VERIFICATION EVIDENCE

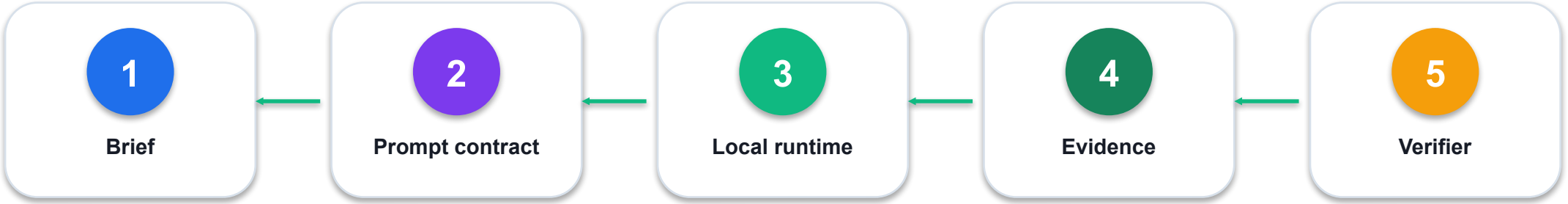
Calls ≤ 40 and triangles ≤ 80,000 for the lab contract; FPS returns above target.

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TOPIC 3 · HANDS-ON LAB · 90 MIN

Lab 08 · Runtime Review, Optimisation and Handover

Review a generated AR/VR runtime, isolate defects with evidence, implement bounded adjustments, and prepare a reproducible handover.

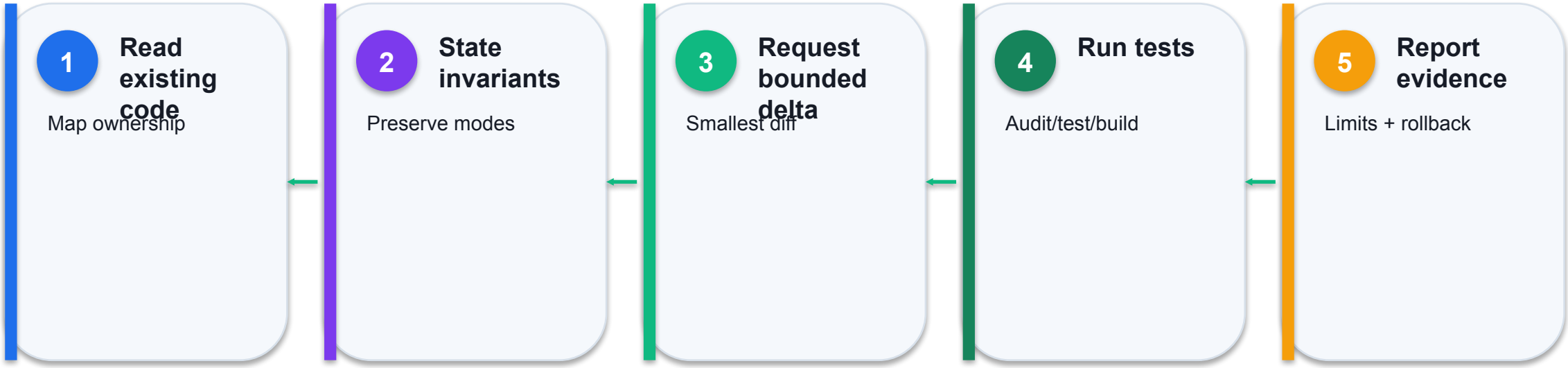


YOU'LL PRODUCE

Scene JSON · runtime evidence · review artifact · PASS Lab 08. Full detailed procedure: labs/lab-08-runtime-review-optimisation-and-handover/README.md

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Lab 08: prompt-to-runtime contract



GUARDRAIL

Generated code remains a proposal until the JSON contract, runtime state and verifier agree.

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Lab 08: technical excerpt

```
{
  "targetMode": "ar",
  "tracking": {
    "plane": "horizontal",
    "placement": "hit-test",
    "anchorPersistence": "session"
  },
  "content": {
    "subject": "building",
    "instances": 5,
    "callouts": 5,
    "scaleMetres": 0.65
  },
  "performance": {
    "targetFps": 60,
    "drawCallBudget": 40,
    "triangleBudget": 80000
  }
}
```

1 **ASSERT 1** all five contract checks pass

2 **ASSERT 2** one defect is evidenced

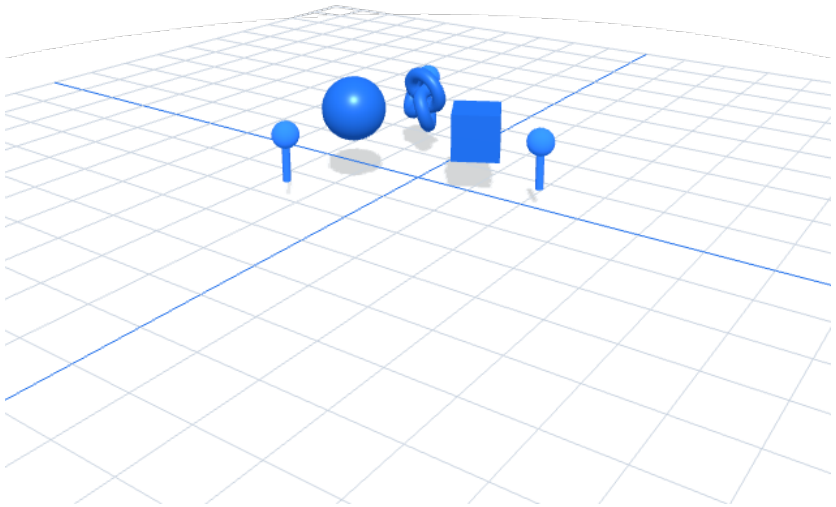
3 **ASSERT 3** one bounded adjustment is retested

4 **ASSERT 4** handover includes commands and rollback

RUN **node labs/lab-08-runtime-review-optimisation-and-handover/verify.mjs → PASS Lab 08**

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Lab 08: acceptance evidence



PASS EVIDENCE

✓ all five contract checks pass

✓ one defect is evidenced

✓ one bounded adjustment is retested

✓ handover includes commands and rollback

DIAGNOSE

- Low FPS: separate device limits from application load before lowering quality.
- Growing memory: verify disposal during scene regeneration.
- Mobile overflow: measure scrollWidth against clientWidth and fix the responsible element, not the whole layout.

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Memory-leak diagnosis

1 Reproduce cycles

Input and state are explicit.

2 Observe counters

Transition has a named trigger.

3 Find retained resource

Output becomes reviewable evidence.

4 Dispose and retest

Acceptance decides the next state.

MECHANISM RULE

Stable scene regeneration should converge rather than monotonically grow geometry/texture/program counters.

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Memory-leak diagnosis: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
console.table(renderer.info.memory);

// acceptance
Counters stabilise across repeated generate/reset cycles.
```

INPUT	Reproduce cycles
STATE	Observe counters
CONTROL	Find retained resource
EVIDENCE	Dispose and retest

ACCEPTANCE — Counters stabilise across repeated generate/reset cycles.

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Memory-leak diagnosis: failure and control

! FAILURE MODE

A one-time warm-up is mistaken for a leak or an actual retained texture is missed.

✓ CONTROL / RECOVERY

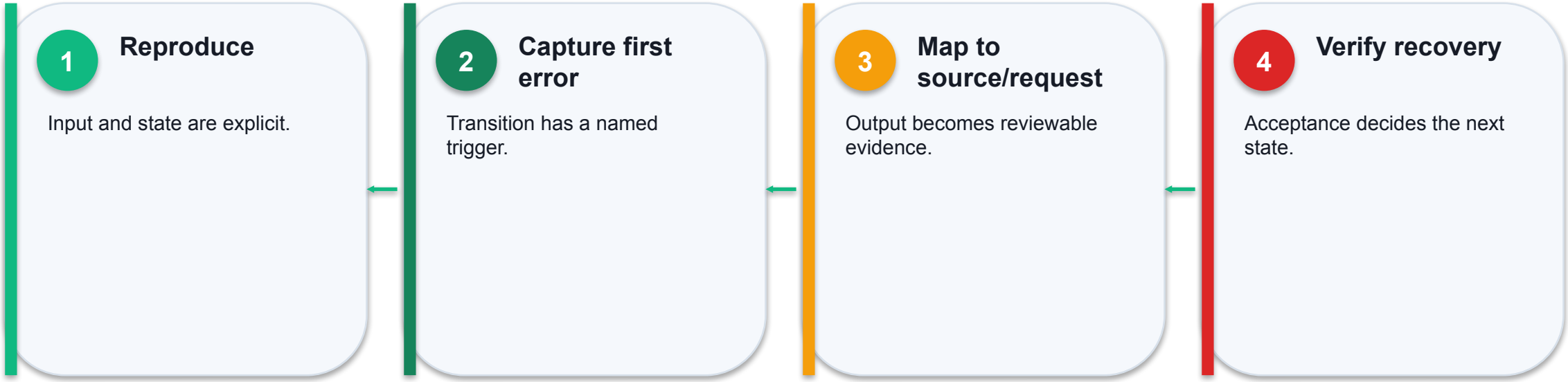
Compare a repeated fixed cycle after warm-up and trace ownership before disposal.

VERIFICATION EVIDENCE

Counters stabilise across repeated generate/reset cycles.

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Console and network evidence



MECHANISM RULE

The first causal error often explains later failures; failed module/asset requests are runtime evidence, not cosmetic warnings.

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Console and network evidence: concrete contract

CONTRACT EXCERPT

// inspectable learner evidence

npm run dev; open DevTools Console and Network

// acceptance

Reload produces zero unexpected console errors and zero failed required requests.

INPUT	Reproduce
STATE	Capture first error
CONTROL	Map to source/request
EVIDENCE	Verify recovery

ACCEPTANCE — **Reload produces zero unexpected console errors and zero failed required requests.**

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Console and network evidence: failure and control

FAILURE MODE

The team fixes the last cascading error or ignores a 404 that removed an AR asset.

CONTROL / RECOVERY

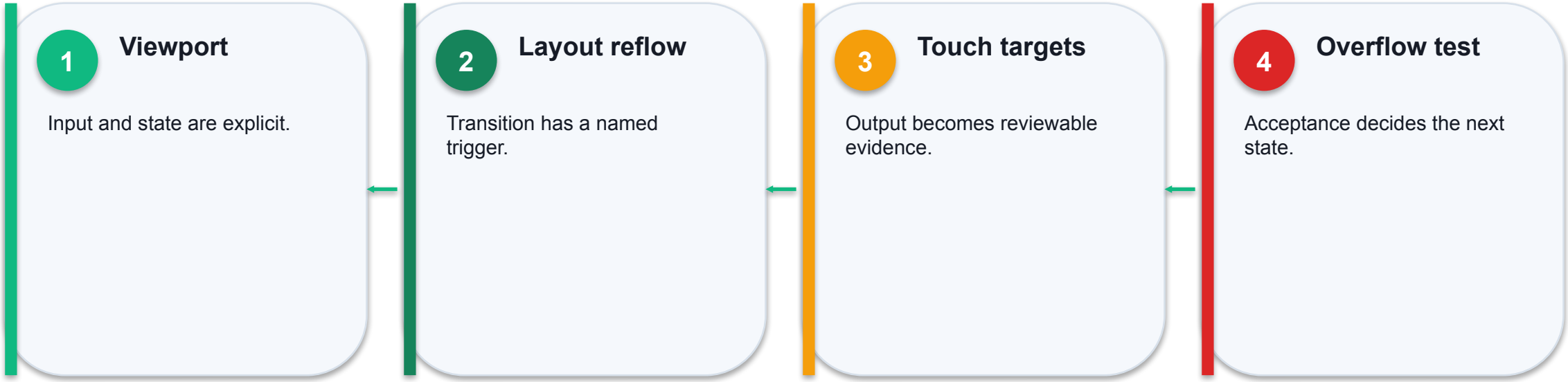
Start from a clean reload, capture timestamps/URL/status and fix the earliest cause.

VERIFICATION EVIDENCE

Reload produces zero unexpected console errors and zero failed required requests.

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Responsive control surface



MECHANISM RULE

Immersive launch and fallback controls must remain usable on the device that hosts the AR session.

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Responsive control surface: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
document.documentElement.scrollWidth <=
document.documentElement.clientWidth

// acceptance
375×667, 390×844 and 768×1024 show no horizontal overflow; actionable
targets meet the touch rule.
```

INPUT	Viewport
STATE	Layout reflow
CONTROL	Touch targets
EVIDENCE	Overflow test

ACCEPTANCE — **375×667, 390×844 and 768×1024 show no horizontal overflow; actionable targets meet the touch rule.**

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Responsive control surface: failure and control

FAILURE MODE

A desktop three-column inspector overflows a 375 px phone or uses sub-44 px controls.

CONTROL / RECOVERY

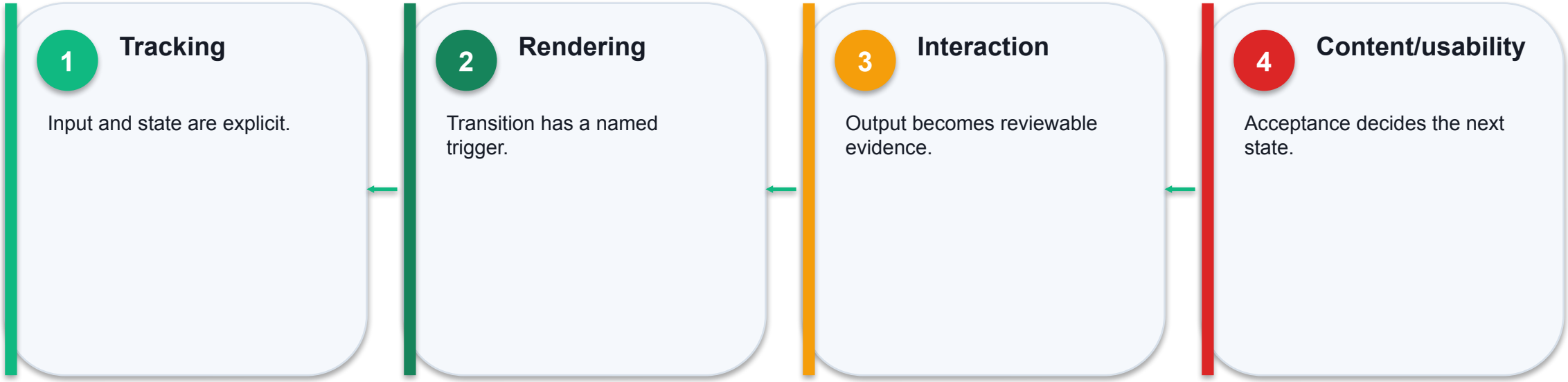
Stack panels, preserve 16 px form text and measure tap targets/overflow.

VERIFICATION EVIDENCE

375×667, 390×844 and 768×1024 show no horizontal overflow; actionable targets meet the touch rule.

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Runtime defect taxonomy



MECHANISM RULE

Classify by observed subsystem before proposing fixes; one symptom may cross boundaries but needs a first testable hypothesis.

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Runtime defect taxonomy: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
defect = {expected, observed, evidence, subsystem, impact}

// acceptance
Every defect record names expected/observed state and a reproducible observation.
```

INPUT	Tracking
STATE	Rendering
CONTROL	Interaction
EVIDENCE	Content/usability

ACCEPTANCE — **Every defect record names expected/observed state and a reproducible observation.**

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Runtime defect taxonomy: failure and control

FAILURE MODE

Drift is blamed on the 3D model when the actual issue is anchor/session lifecycle.

CONTROL / RECOVERY

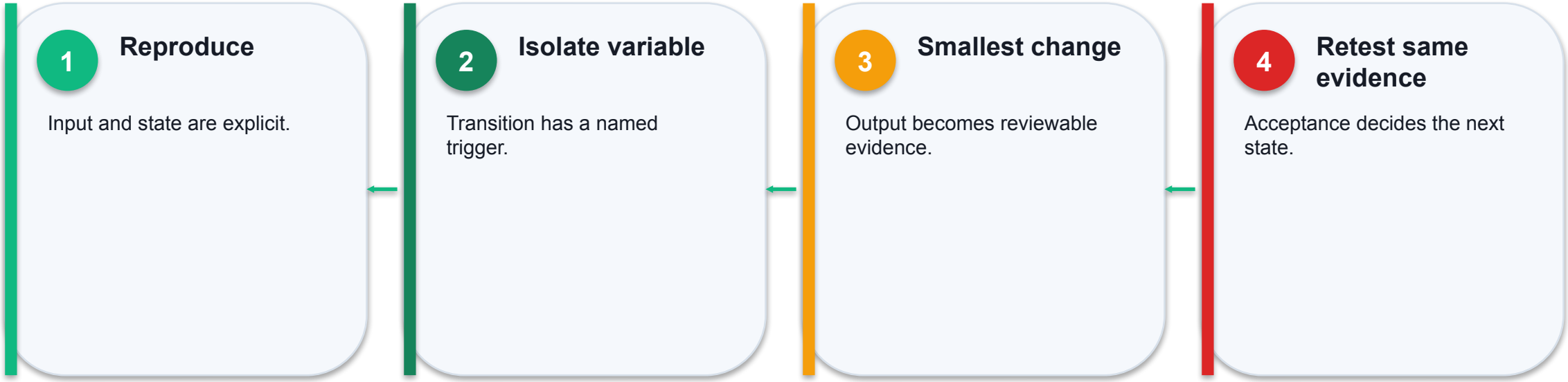
Use subsystem evidence and rank user impact before editing.

VERIFICATION EVIDENCE

Every defect record names expected/observed state and a reproducible observation.

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Root-cause review loop



MECHANISM RULE

A controlled review preserves the failing observation and changes one plausible cause at a time.

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Root-cause review loop: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
test('reticle hides when no hit results', ...)

// acceptance
The original failing case passes and adjacent tests remain unchanged.
```

INPUT	Reproduce
STATE	Isolate variable
CONTROL	Smallest change
EVIDENCE	Retest same evidence

ACCEPTANCE — The original failing case passes and adjacent tests remain unchanged.

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Root-cause review loop: failure and control

FAILURE MODE

A broad rewrite removes the symptom but also changes behaviour and erases causality.

CONTROL / RECOVERY

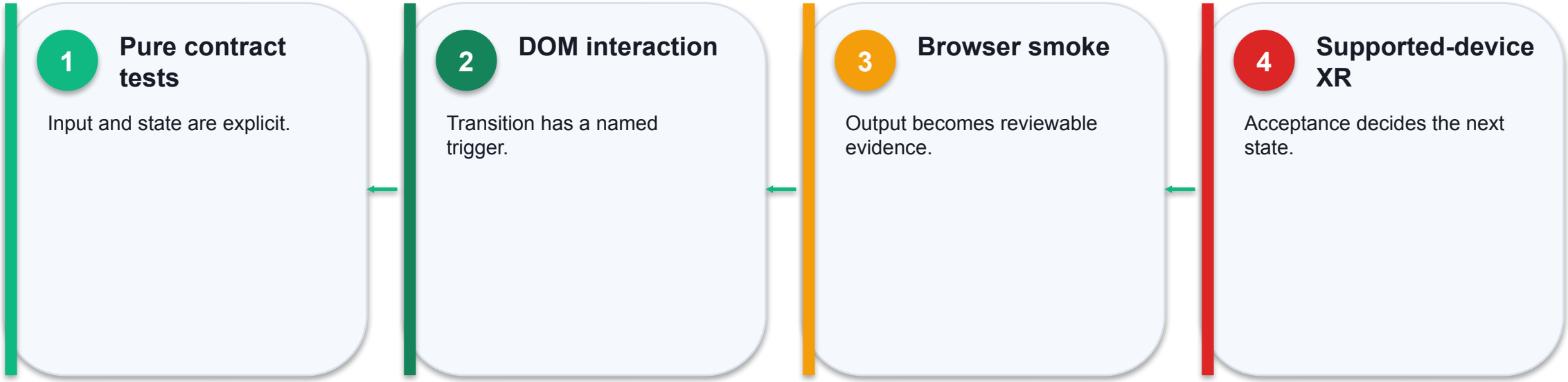
Add a regression test before or with the bounded fix; keep a rollback path.

VERIFICATION EVIDENCE

The original failing case passes and adjacent tests remain unchanged.

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Regression-test pyramid



MECHANISM RULE

Fast deterministic tests cover parsing/validation; browser tests cover UI; physical XR evidence covers device-only behaviour.

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Regression-test pyramid: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
vitest run && vite build

// acceptance
Automated and device-dependent results are reported without conflation.
```

INPUT	Pure contract tests
STATE	DOM interaction
CONTROL	Browser smoke
EVIDENCE	Supported-device XR

ACCEPTANCE — Automated and device-dependent results are reported without conflation.

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Regression-test pyramid: failure and control

FAILURE MODE

A unit test is presented as proof that hit testing worked on a real device.

CONTROL / RECOVERY

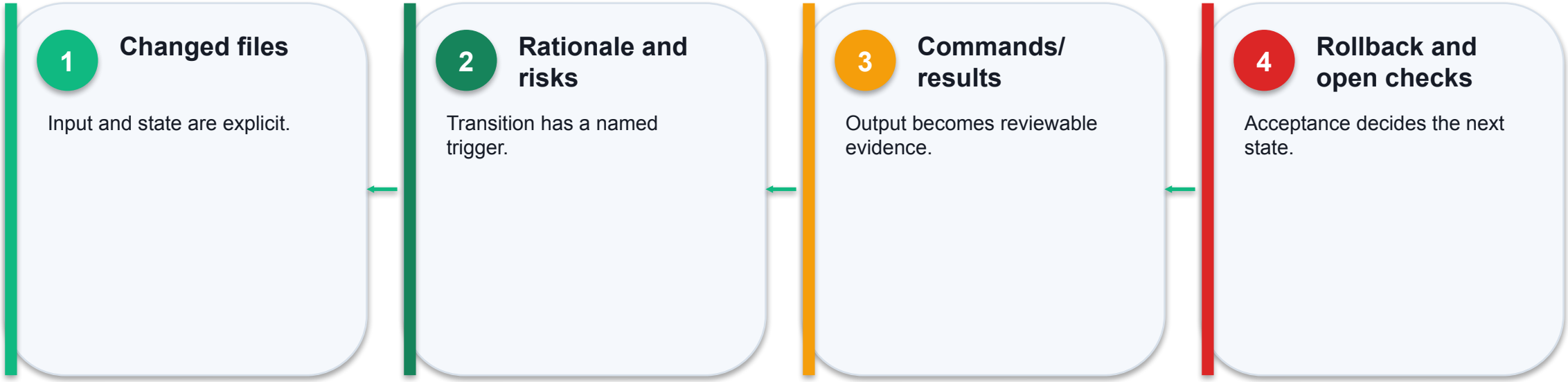
Name the evidence boundary and list device checks separately.

VERIFICATION EVIDENCE

Automated and device-dependent results are reported without conflation.

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Developer handover contract



MECHANISM RULE

Handover records the exact build, scene schema, constraints, evidence and unverified device assumptions.

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Developer handover contract: concrete contract

CONTRACT EXCERPT

```
// inspectable learner evidence
git diff --check; npm audit; npm test; npm run build

// acceptance
A new developer can clone, install, run, verify and revert without hidden context.
```

INPUT	Changed files
STATE	Rationale and risks
CONTROL	Commands/results
EVIDENCE	Rollback and open checks

ACCEPTANCE — **A new developer can clone, install, run, verify and revert without hidden context.**

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Developer handover contract: failure and control

FAILURE MODE

A handover says 'working' without version, tests, or supported-device scope.

CONTROL / RECOVERY

Use a reproducible checklist and link evidence artifacts by relative path.

VERIFICATION EVIDENCE

A new developer can clone, install, run, verify and revert without hidden context.

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REQUIREMENTS → RUNTIME → REVIEW

Course synthesis

1 TRANSLATE

Turn work and equipment facts into a versioned scene contract.

2 STORYBOARD

Model Entry, Orient, Act and Confirm as observable states.

3 BUILD

Run one scene graph through desktop, AR and VR capability boundaries.

4 REVIEW

Diagnose evidence, change one cause, retest and hand over.

COURSE RULE

The coding agent accelerates implementation; the learner owns the contract and evidence.

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CONTINUE YOUR LEARNING

Recommended next courses

1 **WEB**

Vibe Coding for Basic Web Design

2 **REACT**

AI Vibe Coding with React

3 **FULL STACK**

AI Vibe Coding for Full Stack Web Applications

4 **API**

Create RESTful APIs and Web Apps with Python Flask

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AFTER CLASS

Support

1 EMAIL

enquiry@tertiaryinfotech.com

2 PHONE

+65 6100 0613

3 COURSE

https://www.tertiarycourses.com.sg/wsq-ai-vibe-coding-for-augmented-reality-ar-applications.html

4 LIVE PLAYGROUND

https://alfredang.github.io/arvr-playground/

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CLOSING 1 OF 3

Assessment reminder

1 **WA**

7 open-ended questions · 60 minutes

2 **PP**

3 connected tasks · 90 minutes

3 **SUBMIT**

Upload completed candidate documents to the LMS.

4 **SIGN-OFF**

Review feedback and sign the Assessment Summary Record.

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Assessment flow



DECISION

Competent requires all criteria; appeal intent is recorded when the candidate disagrees.

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Digital Attendance (Mandatory)

1 SCAN

Use the live SSG QR displayed by the trainer.

2 SUBMIT

Complete the assessment attendance event.

3 CONFIRM

Check that the trainer sees the attendance status.

4 PROCEED

Begin the assessment only after the briefing and attendance step.

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THANK YOU

Build the scene.
Inspect the contract.
Prove the runtime.

Questions · reflections · next experiment

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